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Elementary Mathematics II MATH 102

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Study Session 1: Function of Real Variable

Introduction

In this session, we will explore the concept of limits. Limits are fundamental tools in calculus used to analyse the behaviour of functions as their inputs approach a particular value. They provide a way to understand what happens to a function near a certain point or as it tends towards infinity.

Learning Outcomes for Study Session 1:

When you have studied this session, you should be able to:

- 1.1. Define and use correctly all of the keywords printed in bold.
- 1.2. Explain the concept of limits and functions and their significance in mathematics.
- 1.3. Identify and analyse the properties of limits, including continuity and one-sided limits.
- 1.4. Calculate the limit of a polynomial function using algebraic techniques.
- 1.5. Apply various methods, such as factoring and rationalisation, to evaluate limits of expressions involving indeterminate forms.

1.1: Limit and Continuous Functions

1.1.1: Informal Definition of Limits

While it is easy to define precisely in a few words what a square root is ($\sqrt{a}$ is the positive number whose square is a) The definition of the limit of a function runs over several terse lines, and most people don't find it very enlightening when they first see it (see figure 1.1). So, we postpone this for a while and fine tune our intuition for another page.

Definition of limit (1st attempt).

If f is some of a function $f(x) = L$

is read "the limit of $f(x)$ as x approaches a is L ". It means that if you choose values of x which are close but not equal to a , then $f(x)$ will be close to the value L ; moreover, $f(x)$ gets closer and closer to L as x gets closer and closer to a .

The following alternative notation is sometimes used

$f(x) \rightarrow L$ as $x \rightarrow a$;

read " $f(x)$ approaches L as x approaches a " or $f(x)$ goes to L as x goes to a ".)

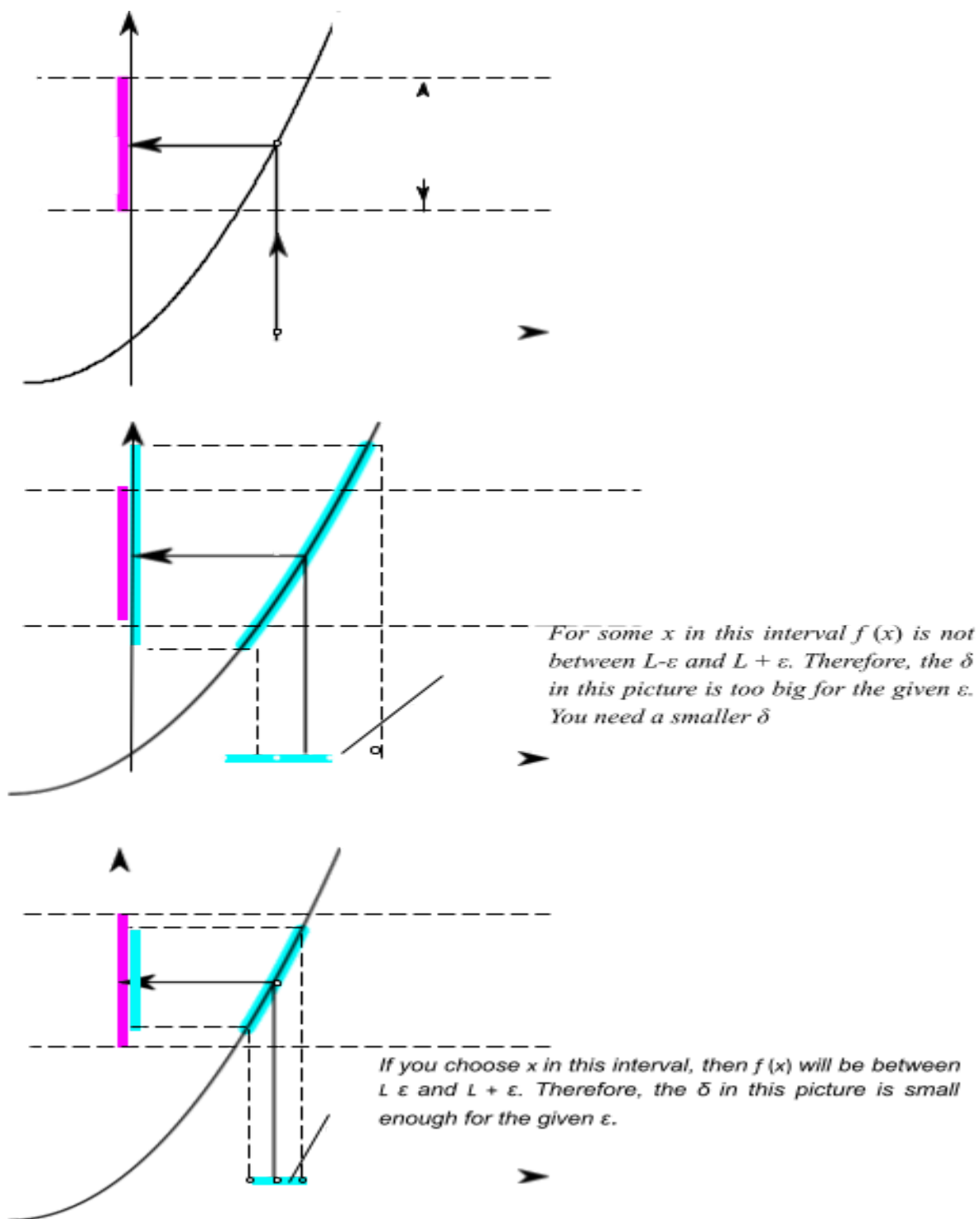


Figure 1.1: Illustrative Diagrams Explaining the Limit of a Function

Example 1. If $f(x) = x + 3$ then

$$f(x) = 7,$$

is true, because if you substitute number x close to 4 in $f(x) = x + 3$ the result will be close to 7.

Example 2. Substituting numbers to guess a limit. What (if anything) is

$$\frac{x^2 - 2x}{x^2 - 4} ?$$

Here, $f(x) = \frac{(x^2 - 2x)}{(x^2 - 4)}$ and $a = 2$.

We first try to substitute $x = 2$, but this leads to

$$f(2) = \frac{2^2 - 2 \cdot 2}{2^2 - 4} = \frac{0}{0}$$

Which does not exist. Next, we try to substitute values of x close but not equal to 2. Table 1.1. suggests that $f(x)$ approaches 0.5

Table 1.1. finding limits by substituting values of x "close to a ". (values of $f(x)$ and $g(x)$ recorded to six decimals.)

x	$f(x)$	x	$g(x)$
3.000000	0.600000	1.000000	1.009990
2.500000	0.555556	0.500000	1.009980
2.100000	0.512195	0.100000	1.009899
2.010000	0.501247	0.010000	1.008991
2.001000	0.500125	0.001000	1.000000

Example 3. Substituting numbers can suggest the wrong answer. The previous example shows that our first definition of “limit” is not very precise, because it says “ x close to a ”, but how close is close enough? Suppose we had taken the function

$$g(x) = \frac{101000x}{100000x+1}$$

and we had asked for the limit $g(x)$.

Then substitution of some “small values of x ” could lead us to believe that the limit is 1.00... only when you substitute even smaller values do you find that the limit is 0 (zero)!

1.1.2: The formal, authoritative, definition of limit.

The formal description of the limit uses phrases like “closer and closer” and “really very small.” In the end we don’t really know what they mean, although they are suggestive. “fortunately” there is a good definition, i.e. one which is unambiguous and can be used to settle any dispute about the question of whether $f(x)$ equals some number L or not. Here is the definition. It takes a while to digest, so read it once, look at the examples, do a few exercises, read the definition again. Go on to the next sections. Throughout the semester come back to this section and read it again.

1.1.3: Definition of Limit of $f(x) = L$.

We say that L is the limit of $f(x)$ as $x \rightarrow a$, if

1. $f(x)$ need not be defined as $x = a$, but it must be defined for all other x in some interval which contain a .
2. For every $\epsilon > 0$ one can find a $\delta > 0$ such that for all x in the domain of f one has

$$|x - a| < \delta \text{ implies } |f(x) - L| < \epsilon.$$

Why the absolute value? The quantity $|x - y|$ is the distance between the points x and y on the number line, and one can measure how close x is to y by calculating $|x - y|$. The inequality $|x - y| < \delta$ says that “the distance between x and y is less than δ ,” or “ x and y are closer than δ .”

Why are ϵ and δ ? The quantity ϵ is how close you would like $f(x)$ to be to its limit L ; the quantity δ is how close you have to choose x to a to achieve this. To prove that $f(x) = L$ you must assume that someone has given you an unknown $\epsilon > 0$, and then find a positive δ for which holds. The δ you find will depend on ϵ .

- (a) **Show that** $2x + 1 = 11$. We have $f(x) = 2x + 1$, $a = 5$ and $L = 11$, and the question we must answer is “how close should x be to 5 if we want to be sure that $f(x) = 2x + 1$ differs less than ϵ from $L = 11$?”

To figure this out we try to get an idea of how big $|f(x) - L|$ is:

$$|f(x) - L| = |(2x + 1) - 11| = |2x - 10| = 2 \cdot |x - 5|.$$

So, if $2|x - 5| < \epsilon$ then we have $|f(x) - L| < \epsilon$, i. e.

$$\text{If } |x - a| < \frac{1}{2}\epsilon \text{ then } |f(x) - L| < \epsilon$$

We can therefore choose $\delta = \frac{1}{2}\epsilon$. No matter what $\epsilon > 0$ we are given our δ will also be positive, and if $|x - 5| < \delta$ then we can guarantee $|(2x + 1) - 11| < \epsilon$.

That shows that $2x + 1 = 11$.

(b) **The limit $x^2 - 1$ and the “don’t choose $\delta > 1$ ” trick.**

We have $f(x) = x^2$, $a = 1$, $L = 1$, and again the question is, “how small should $|x - 1|$ be guarantee $|x^2 - 1| < \epsilon$?”

we begin by estimating the difference $|x - 1|$

$$|x^2 - 1| = |(x - 1)(x + 1)| = |x + 1| \cdot |x - 1|.$$

As x approaches 1 the factor $|x - 1|$ becomes small, and if the other factor $|x + 1|$ were a constant (*e. g. 2 as in the previous example*) then we could find δ as before, by dividing ϵ by that constant.

Here is a trick that allows you to replace the factor $|x + 1|$ with a constant. We hereby agree that we always choose our δ so that $\delta \leq 1$. If we do that, then we will always have

$$|x^2 - 1| = |x + 1| \cdot |x - 1| < 3|x - 1| < 1.$$

and x will always be between 0 and 2. Therefore

$$|x^2 - 1| = |x + 1| \cdot |x - 1| < 3|x - 1| < 1$$

If we now want to be sure that $|x^2 - 1| < \epsilon$, then this calculation shows that we should require $3|x - 1| < \epsilon$, *i. e.* $|x - 1| < \frac{1}{3}\epsilon$. So we should choose $\delta \leq \frac{1}{3}\epsilon$. We must also live up to our promise never to choose $\delta > 1$, so if we are handed an

ϵ for which $\frac{1}{3}\epsilon > 1$, then we choose $\delta = 1$ instead of $\delta = \frac{1}{3}\epsilon$. To summarise, we are going to choose.

$$\delta = \text{the smaller of } 1 \text{ and } \frac{1}{3}\epsilon.$$

We have shown that if you choose δ this way, then $|x - 1| < \delta$ implies $|x^2 - 1| < \epsilon$, no matter what $\epsilon > 0$ is.

The expression “the smaller of a and b ” shows up often, and is abbreviated to (a, b) . we could therefore say that in this problem we will choose δ to be

$$\delta = \left(1, \frac{1}{3}\epsilon\right).$$

(c) **Show that** $\frac{1}{x} = \frac{1}{4}$. Solution: we apply the definition with

$a = 4$. $L = \frac{1}{4}$ and $f(x) = \frac{1}{x}$. Thus, for any $\epsilon > 0$ we try to show that

$|x - 4|$ is small enough then one has $\left|f(x) - \frac{1}{4}\right|$ in terms of $|x - 4|$:

$$\left|f(x) - \frac{1}{4}\right| = \left|\frac{1}{x} - \frac{1}{4}\right| = \left|\frac{4-x}{4x}\right| = \frac{|x-4|}{|4x|} = \frac{1}{|4x|}|x - 4|.$$

As before, things would be easier if $\frac{1}{|4x|}$ were a constant. To achieve that we again agree not to take $\delta > 1$. If we always have $\delta \leq 1$, then we will always have $|x - 4| < 1$, and hence $3 < x < 5$. How large can $\frac{1}{|4x|}$ be in this situation?

Answer: the quantity $\frac{1}{|4x|}$ increase x , so if $3 < x < 5$ then it will never be large than $\frac{1}{|4 \cdot 3|} = \frac{1}{12}$.

We see that we never choose $\delta > 1$, we will always have

$$\left|f(x) - \frac{1}{4}\right| \leq \frac{1}{12}|x - 4| \text{ for } |x - 4| < \delta.$$

To guarantee that $\left|f(x) - \frac{1}{4}\right| < \epsilon$ we could therefore require

$$\frac{1}{12}|x - 4| < \epsilon, \text{ i. e. } |x - 4| < 12\epsilon.$$

Hence if we choose $\delta = 12\epsilon$ or any smaller number, then $|x - 4| < \delta$ implies $|f(x) - \frac{1}{4}| < \epsilon$ of course we have to honour our agreement never to choose $\delta > 1$, so our choice of δ is

$$\delta = \text{the smaller of } 1 \text{ and } 12\epsilon = (1, 12\epsilon)$$

Exercises:

1. Group problem

Joe offers to make square sheets of paper for Bruce. Given $x > 0$ Joe plans to make off a length x and cut out a square of side x . Bruce asks Joe for a square with area 4 square foot. Joe tells Bruce that he can't measure exactly 2 foot and the area of the square he produces will only be approximately 4 square foot. Bruce doesn't mind as long as the area of the square doesn't differ more than 0.01 square foot from what he really asked for (namely, 4 square foot).

- What is the biggest error Joe can afford to make when he marks off the length x ?
- Jen also wants square sheets, with area 4 square feet. However, she needs the error in the area to be less than 0.00001 square foot. (she's paying).

How accurate must Joe measure the side of the squares he's going to cut for Jen?

Use the $\epsilon - \delta$ definition to prove the following limits

2. $2x - 4 = 6$

3. $x^2 = 4$

4. $x^2 - 7x + 3 = -7$

5. $x^3 = 27$

6. $x^3 + 6x^2 = 32$

7. $\sqrt{x} = 2$

8. $\sqrt{x + 6} = 9$

9. $\frac{1+x}{4+1} = \frac{1}{2}$

10. $\frac{2-x}{4-x} = \frac{1}{3}$

11. $\frac{x}{6-x} = 1$

12. $\sqrt{|x|} = 0$

13. **Group problem**

(Joe goes cubic) Joe is offering to build cubes of side x . Airlines regulations allow you take a cube on board provided its volume and surface area add up to less than 33 (everything measured in feet). For instance, a cube with 2 foot sides has volume + area to $2^3 + 6 + 2^2 = 32$.

If you ask Joe to build a cube whose volume plus total surface area is 32 cubic feet with an error of at most ϵ , then what error can he afford to make when he measures the side of the cube he's making?

14. our definition of a derivative in (7) contains a limit. What is the function " f " there, and what is the variable?

Note: Variations on the limit theme

Not all limits are " $\text{for } x \rightarrow a$," here we describe some possible variations on the concept of limit.

1.1.4: Left and right limit.

When we let " x approach a " we allow x to be both larger or smaller than a , as long as x gets close to a . If we explicitly want to study the behaviour of $f(x)$ as x approaches a through values larger than a , then we write

$$f(x) \text{ or } f(x) \text{ or } f(x) \text{ or } f(x)$$

All four notations are in use. Similarly, to designate the value which $f(x)$ approaches as x approaches a through values below a one writes

$$f(x) \text{ or } f(x) \text{ or } f(x) \text{ or } f(x)$$

The precise definition of right limits goes like this:

1.1.5: Definition of right-limits.

Let f be a function. Then

$$\lim_{x \rightarrow a^+} f(x) = L \quad (1)$$

This means that for all every $\varepsilon > 0$ one can find a $\delta > 0$ such that

$$a < x < a + \delta \Rightarrow |f(x) - L| < \varepsilon$$

Holds for all x in the domain of f .

The left-limit, i.e., the one-sided limit in which x approaches a through values less than a is defined in a similar way. The following theorem tells you how to use one-sided limits to decide if a function $f(x)$ has a limit at $x = a$.

1.1.6: Theorem. If both are one-sided limits

$$\lim_{x \rightarrow a^-} f(x) = L^+ \text{ and } \lim_{x \rightarrow a^+} f(x) = L^-$$

exists, then

$$\lim_{x \rightarrow a} f(x) \text{ exists} \Leftrightarrow L^+ = L^-$$

In other words, if a function has both left and right-limits at some $x = a$, then function has a limit at $x = a$ if the left and right limits are equal.

Limits at infinity. Instead of letting x approach some finite number, one can let x become “larger and larger” and ask what happens to $f(x)$. If there is a number L such that $f(x)$ gets arbitrary close to L if one chooses x sufficiently large, then we write

$$\lim_{x \rightarrow \infty} f(x) = L \text{ or } \lim_{x \rightarrow -\infty} f(x) = L$$

(“The limit of x going to infinity is L ”)

Example 4. Limit of $\frac{1}{x}$. the larger you choose x , the smaller its reciprocal $\frac{1}{x}$ becomes. Therefore, it seems reasonable to say

$$\frac{1}{x} = 0$$

Here is the precise definition:

1.1.7: Definition of limit at Infinity ∞ .

Let f be some function which is defined on some interval $x_0 < x < \infty$. If there is a number L such that for every $\varepsilon > 0$ one can find on A such that

$$x > A \Rightarrow |f(x) - L| < \varepsilon$$

For all x , then we say that the limit of $f(x)$ for $x \rightarrow \infty$ is L .

The definition is very similar to the original definition of the limit. Instead of δ which specifies how close x should be to a , we now have a number A which says how large x should be, which is a way of saying “how close x should be to infinity”

Example 5. Limit of $\frac{1}{x}$ (Again). To prove that $\frac{1}{x} = 0$ we apply the definition to

$$f(x) = \frac{1}{x}, L = 0$$

For given $\varepsilon > 0$ we need to show that

$$\left| \frac{1}{x} - L \right| < \varepsilon \text{ for all } x > A \quad (2)$$

provided we choose the right A .

How do we choose the right A is not allowed to depend on x , but it may depend on ε . If we assume for now that we will only consider positive values of x , then (2) simplifies to

$$\frac{1}{x} < \varepsilon$$

which is equivalent to

$$\varepsilon > \frac{1}{x}$$

This tells us how to choose A . Given any positive ε , we will simply choose

$$A = \frac{1}{\varepsilon}$$

Then one has $\left| \frac{1}{x} - 0 \right| = \frac{1}{x} < \varepsilon$ for all $x > A$. Hence we have proved that

$$\frac{1}{x} = 0$$

1.2: Properties of the Limit

The precise definition of the limit is not easy to use, and fortunate we won't use it very often in this class. Instead, there are a number of properties that limits have which allow you to compute them without having to resort to "epsilon-ness."

The following properties also apply to the variations on the limit from 4. *I. e.* the following statements remain true if one replaces each limit by a one-sided limit, or a limit for $x \rightarrow \infty$.

1.2.1: Limits of constants and of x .

If a and c are constants, then

$$C = C \quad (P_1)$$

$$x = a \quad (P_2)$$

1.2.2 Limits of sums, products and quotients.

Let F_1 and F_2 be two given functions whose limits for $x \rightarrow a$ we now,

$$F_1(x) = L_1, \quad F_2(x) = L_2$$

then

$$(F_1(x) + F_2(x)) = L_1 + L_2 \quad (P_3)$$

$$(F_1(x) - F_2(x)) = L_1 - L_2 \quad (P_4)$$

$$(F_1(x) \cdot F_2(x)) = L_1 \cdot L_2 \quad (P_5)$$

finally, if $\lim_{x \rightarrow a} F_2(x) \neq 0$,

$$\frac{F_1(x)}{F_2(x)} = \frac{L_1}{L_2} \quad (P_6)$$

In other words, the limit of the sum is the sum of the limits, etc. one can prove these laws using the definition of limit in P_2 but we will not do this here.

However, I hope these laws seem like common sense. If for x close a , the quantity $F_1(x)$ is closer to L_1 and $F_2(x)$ is closer to L_2 , then certainly $F_1(x) + F_2(x)$ should be close to $L_1 + L_2$.

There are two more properties of limits which we will add to this list later on. They are the “sandwich theorem” (9) and the substitution theorem (10).

Example 6. Limit computation

(i) Find x^2 . One has

$$\begin{aligned} x^2 &= x \cdot x \\ &= (x) \cdot (x) && \text{by } (P_5) \\ &= 2 \cdot 2 = 4 \end{aligned}$$

similarly,

$$\begin{aligned} x^3 &= x \cdot x^2 \\ &= (x) \cdot (x) && \text{by } (P_5) \text{ again} \\ &= 2 \cdot 4 = 8 \end{aligned}$$

and, by (P_4)

$$x^2 - 1 = x^2 - 1 = 4 - 1 = 3,$$

and, by (P_4) again,

$$x^3 - 1 = x^3 - 1 = 8 - 1 = 7,$$

putting all this together, one gets

$$\frac{x^3-1}{x^2-1} = \frac{2^3-1}{2^2-1} = \frac{8-1}{4-1} = \frac{7}{3}$$

Because of (P_6) . To apply (P_6) we must check that the denominator (L_2) is not zero. Since the denominator is 3 everything is ok, and we were allowed to use (P_6)

(ii) Try examples 2 and 3 using the limit properties. To compute $\frac{(x^2-2x)}{x^2-4}$ we

first use the limit properties to find

$$x^2 - 2x = 0 \text{ and } x^2 - 4 = 0$$

To complete the computation, we would like to apply the last property (P_6) about quotients, but this would give us

$$f(x) = \frac{0}{0}$$

The denominator is zero, so we were not allowed to use (P_6) (and the result doesn't mean anything anyways). We have to do something else.

The function we are dealing with is a rational function, which means that it is the quotient of two polynomials. For such functions there is an algebra trick which always allows you to compute the limit even if you first get $\frac{0}{0}$. The thing to do is to divide numerator and denominator by $x - 2$. In our case we have

$$x^2 - 2x = (x - 2) \cdot x, \quad x^2 - 4 = (x - 2) \cdot (x + 2)$$

so that,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{(x-2) \cdot x}{(x-2) \cdot (x+2)} = \lim_{x \rightarrow 2} \frac{x}{x+2}$$

after this simplification we can use the properties to compute

$$\lim_{x \rightarrow 2} f(x) = \frac{2}{2+2} = \frac{1}{2}.$$

(iii) Example. Find $\lim_{x \rightarrow 2} \sqrt{x}$ of course, you would think that $\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$ and you can indeed prove this using δ and ε (see problem 7). But is there an easier way? There is nothing in the limit properties which tells us how to deal with a square root, and using them we can't even prove that there is a limit. However,

if you assume that the limit exists then the limit properties allow us to find this limit.

The argument goes like this: suppose that there is a number L with.

$$\lim_{x \rightarrow 2} \sqrt{x} = L$$

Then property (P_5) implies that

$$L^2 = \left(\lim_{x \rightarrow 2} \sqrt{x} \right) \cdot \left(\lim_{x \rightarrow 2} \sqrt{x} \right) = \lim_{x \rightarrow 2} \sqrt{x} \cdot \sqrt{x} = \lim_{x \rightarrow 2} x = 2$$

In other words, $L^2 = 2$, and hence L must be either $\sqrt{2} = \sqrt{2}$ or $-\sqrt{2}$. We can reject the latter because whatever x does, its square root is always a positive number, and hence it can never “get close to” a negative number like $-\sqrt{2}$.

Our conclusion: if the limit exists, then

$$\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$$

The result is not surprising: if x get close to 2 then $\sqrt{x}$ gets close to $\sqrt{2}$

(iv) Example- The derivative of $\sqrt{x}$ at $x = 2$. Find

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$$

Assuming the result from the previous example.

Solution: The function is a fraction whose numerator and vanish when $x = 2$.

i.e. The limit is of the form $\frac{0}{0}$. We use the same algebra trick as before, namely

we factor numerator and denominator:

$$\frac{\sqrt{x} - \sqrt{2}}{x - 2} = \frac{\sqrt{x} - \sqrt{2}}{(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})} = \frac{1}{\sqrt{x} + \sqrt{2}}$$

Now one can use the limit properties to compute

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} = \lim_{x \rightarrow 2} \frac{1}{\sqrt{x} + \sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$$

(v) Limit as $x \rightarrow \infty$ of rational function. A rational function is the quotient of two polynomials, so

$$R(x) = \frac{a_n x^n + \dots + a_1 x + a_0}{b_m x^m + \dots + b_1 x + b_0} \quad (3)$$

We have seen that:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

we even proved this in example 4.7. using this you can find the limit at ∞ any rational function $R(x)$ as in (3). One could turn the outcome of the calculation of $R(x)$ into a recipe/formula involving the degrees n and m of the numerator and denominator, and also their coefficients a_i, b_i , which students would then memorise, but it is better to remember “the trick.”

To find $R(x)$ divide numerator and denominator by x^m (the highest power of x occurring in the denominator).

For example, let's compute

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 3}{5x^2 + 7x - 39}$$

remember the trick and divide and bottom by x^2 , and you get

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 3}{5x^2 + 7x - 39} = \lim_{x \rightarrow \infty} \frac{\frac{3+3}{x^2}}{\frac{5+7}{x} - \frac{39}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3+3}{x^2}}{\frac{5+7}{\frac{x-39}{x^2}}}$$

$$= \frac{3}{6}$$

Here we have used the limit properties P_* to break the limit down into little pieces' like $\lim_{x \rightarrow \infty} \frac{39}{x^2}$ which we can compute as follows

$$\lim_{x \rightarrow \infty} \frac{39}{x^2} = \lim_{x \rightarrow \infty} 39 \cdot \left(\frac{1}{x}\right)^2 = \left(\lim_{x \rightarrow \infty} 39\right) \cdot \left(\lim_{x \rightarrow \infty} \frac{1}{x}\right)^2 = 39 \cdot 0^2 = 0$$

(vi) Another example with a rational function. Compute

$$\lim_{x \rightarrow \infty} \frac{x}{x^3 + 5}$$

We apply “the trick” again and divide numerator and denominator by x^3 . This leads to

$$\lim_{x \rightarrow \infty} \frac{x}{x^3 + 5} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}}{\frac{x^3 + 5}{x^3}} = \frac{0}{1} = 0$$

To show all possible ways a limit of a rational function can turn out we should do yet another example, but that one belongs in the next section.

When limit fail to exist

In the last couple of examples, we worried about the possibility that a limit $\lim_{x \rightarrow \infty} g(x)$ actually might not exist. This can actually happen, and in this section, we will see a few examples of what failed limits look like. First let's agree on what we will call a “failed limit.”

1.9 Definition. If there is no number L such that $\lim_{x \rightarrow \infty} f(x) = L$, then we say that the limit $\lim_{x \rightarrow \infty} f(x) = L$ does not exist.

1.10 The sign function near $x = 0$.

The “the function” is defined by

$$\text{sign}(x) = \begin{cases} -1 & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ 1 & \text{for } x > 0 \end{cases}$$

Note that “the sign function” is not to be zero. But does the sign function have a limit at $x = 0$, i.e. does $\lim_{x \rightarrow 0} \text{sign}(x)$ exist? And is it also zero? The answers are **no**, and **no** and here is why: suppose that for some number L one had.

$$\lim_{x \rightarrow 0} \text{sign}(x) = L$$

Then since for arbitrary small positive values of x one has $\text{sign}(x) = +1$ one would think that $L = +1$. But for arbitrarily small negative values of x one has $\text{sign}(x) = -1$, so one could conclude that $L = -1$. But one number L can't be both $+1$ and -1 at the same time, so there is no such L , i.e., there is no limit.

$$\lim_{x \rightarrow 0} \text{sign}(x) \text{ does not exist}$$

Some people don't like the notation $\text{sign}(x)$, and prefer to write

$$g(x) = \frac{x}{|x|}$$

instead of $g(x) = \text{sign}(x)$. if you think about this formula for a moment you will see that $\text{sign}(x) = \frac{x}{|x|}$ for all $x \neq 0$. When $x = 0$ the quotient $\frac{x}{|x|}$ is of course not defined.

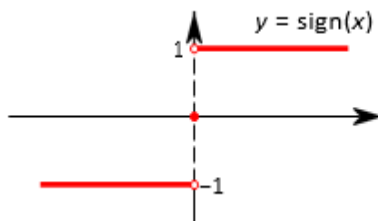


Figure 1.2: The sign function.

In this example the one-sided limit does exist, namely,

$$\lim_{x \searrow 0} \text{sign}(x) = 1 \text{ and } \lim_{x \nearrow 0} \text{sign}(x) = -1$$

All this says is that when x approaches 0 through positive values, its sign approaches + 1, while if x goes to 0 through negative values, then its sign approaches – 1.

The example of the backward sine.

Contemplate the limit as $x \rightarrow 0$ of the “backward sine.”

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$$

when $x = 0$ the function $f(x) = \sin\left(\frac{\pi}{x}\right)$ is not defined, because its definition involves division by x . What happens to $f(x)$ as $x \rightarrow 0$? First, $\frac{\pi}{x}$ become larger and larger (“goes to infinity”) as $x \rightarrow 0$. Then, that the sine, we see that $\sin\left(\frac{\pi}{x}\right)$

oscillates between $+1$ and -1 infinitely often as $x \rightarrow 0$. This means that $f(x)$ gets close to any number between $+1$ and -1 as $x \rightarrow 0$, but that the function $f(x)$ never stays close to any particular value because it keeps oscillating and up and down.

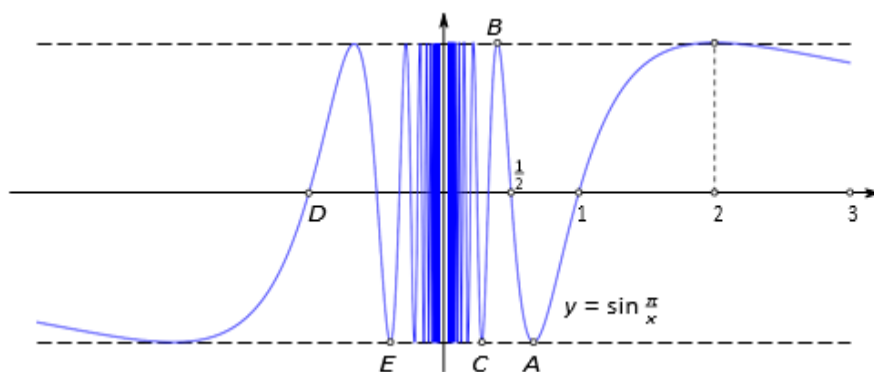


Figure 1.3: Graph of $y = \sin \frac{\pi}{x}$ for $-3 < x < 3, x \neq 0$

Here again, the limit $\lim_{x \rightarrow 0} f(x)$ does not exist. We have arrived at this conclusion by only considering what $f(x)$ does for small positive values of x . So, the limit fails to exist in a stronger way than in the example of the sign-function. There, even though the limit didn't exist, the one-sided limits existed. In the present example we see that even the one-sided limit

$$\lim_{x \searrow 0} \sin \frac{\pi}{x}$$

Does not exist.

Dividing by zero using a limit.

The expression $\frac{1}{0}$ is not defined, but what about

$$\lim_{x \rightarrow 0} \frac{1}{x}?$$

This limit also does not exist. Here are two reasons:

It is common wisdom that if you divide by a small number, you get a large number, so as $x \searrow 0$ the quotient $\frac{1}{x}$ will not be able to stay close to any particular finite number, and the limit can't exist.

“Common wisdom” is not always a reliable tool in mathematical proofs, so here is a better argument. The limit can't exist, because that would contradict the limit properties $(P_1) \dots (P_6)$. Namely, suppose that there were a number L such that

$$\lim_{x \rightarrow 0} \frac{1}{x} = L$$

Can't exist, for if it did, then limit

$$\lim_{x \rightarrow 0} \frac{1}{x} = \lim_{x \rightarrow 0} \left(\frac{1}{x} - x + x \right) = \lim_{x \rightarrow 0} \left(\frac{1}{x} - x \right) + \lim_{x \rightarrow 0} x$$

Would also have to exist, and we now $\lim_{x \rightarrow 0} \frac{1}{x}$ doesn't exist.

Limits at ∞ which don't exist.

If you let x go to ∞ , then x will not get “closer and closer” to any particular number L , so it seems reasonable to guess that

x does not exist

One can prove this from the limit definition (and see exercise 72)

Let's consider

$$L = \lim_{x \rightarrow \infty} \frac{x^2 + 2x - 1}{x}$$

Here, the denominator has a limit ('tis 1), but the numerator does not, for if $x^2 + 2x - \frac{1}{x}$ existed then, since $\left(\frac{x+2}{x}\right) = 2$ exist.

$$x = \left[\left(x^2 + 2x - \frac{1}{x} \right) - \left(2 - \frac{1}{x} \right) \right]$$

Would also have to exist, and x doesn't exist.

So, we see that L is the limit of a fraction in which the denominator has a limit, but the numerator does not. In this situation the limit L itself can never exist. If it did, then

$$\left(x + 2 - \frac{1}{x} \right) = \lim_{x \rightarrow \infty} \frac{x+2-\frac{1}{x}}{\frac{x+2}{x}} \cdot \left(\frac{1+2}{x} \right)$$

Would also have to have a limit.

What's in a name?

There is a big difference between the variable x and a in the formula

$$\lim_{x \rightarrow a} 2x + 1,$$

Namely a is a **free variable**, while x is a **dummy variable** (or “placeholder” or a “bound variable”)

The difference between these two kinds of variable is this:

- If you replace a dummy variable in some formula consistently by some other variable, then the value of the formula does not change. On the other, it never makes sense to substitute a number for a dummy variable
- The value of the formula may depend on the value of the free variable.

To understand what this means consider the example $\lim_{x \rightarrow a} 2x + 1$ again. The limit is easy to compute:

$$\lim_{x \rightarrow a} 2x + 1 = 2a + 1$$

If we replace x by, say u (systematically) then we get

$$\lim_{x \rightarrow a} 2u + 1$$

Which is again equal to $2a + 1$. This computation says that if some number gets close to a then two times that number plus one gets close to $2a + 1$. This is a very wordy way of expressing the formula, and you can shorten things giving a name (like x or u) to the number which approaches a . But the result of our computation shouldn't depend on the name choose, i.e. it doesn't matter if we call it x or u .

Since the name of the variable x doesn't matter, it is called a dummy variable. Some prefer to call x a bound variable, meaning that in

$$\lim_{x \rightarrow a} 2x + 1$$

The x in the expression $2x + 1$ is bound to the x written underneath the limit-you can't change one without changing the other.

Substituting a number for a dummy variable usually leads to complete nonsense. For instance, let's try setting $x = 3$ in our limit, i.e. what is

$$\lim_{3 \rightarrow a} 2.3 + 1?$$

Of course, $2.3 + 1 = 7$, but what does 7 do when 3 get closer to the number a ? That's is silly question, because 3 is a constant and it doesn't "get closer" to

some other number like a ! If you ever see 3 get closer to another number the it's time to take a vacation.

On the other hand, the variable a is free: you can assign it particular values, and its value will affect the value of the limit. For instance, if we set $a = 3$ (but

leave x alone) then we get $\lim_{x \rightarrow 3} 2x + 1$

and there's nothing strange about that (the limit is $2 \cdot 3 + 1 = 7$, no problem). You could substitute other value of a and you would get a different answer. In general, you get $2a + 1$.

1.3: Limits and inequalities

This section has two theorems which let you compare limits of different functions. The properties in these theorems are not formulas that allow you to compute limits like the properties $(P_1) \dots (P_6)$. Instead, they allow you to reason about limits, i.e., they let you know that this or that limit is positive, or that it must be the same as some other limit which you find easier to think about.

The first theorem should not surprise you-all its say is that bigger functions have bigger limits.

1.3.1: Theorem: Limits and Inequalities for Functions

Let f and g be functions whose limits for $x \rightarrow a$ exist, and assume that $f(x) \leq g(x)$ holds for all x . Then

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$$

A usual special case arises when you set $f(x) = 0$. The theorem then says that if a function g never has negative values, then its limits will also never be negative.

The statement may seem obvious, but it still needs a proof, starting from the $\varepsilon - \delta$ definition of limit. This will be done in lecture.

Here is the second theorem about limits and inequalities.

1.3.2: The Sandwich Theorem.

Suppose that

$$f(x) \leq g(x) \leq h(x)$$

(for all x) and that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$$

then

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x)$$

The theorem is useful when you want to know the limit of g , and when you can **sandwich** it between two functions f and h whose limits are easier to compute. The sandwich Theorem looks like the first theorem of this section, but there is an important difference. In the Sandwich Theorem you don't have to assume that the limit of g exists. The inequalities $f \leq g \leq h$ combined with the circumstance that f and h have the same limit are enough to guarantee that the g exists.

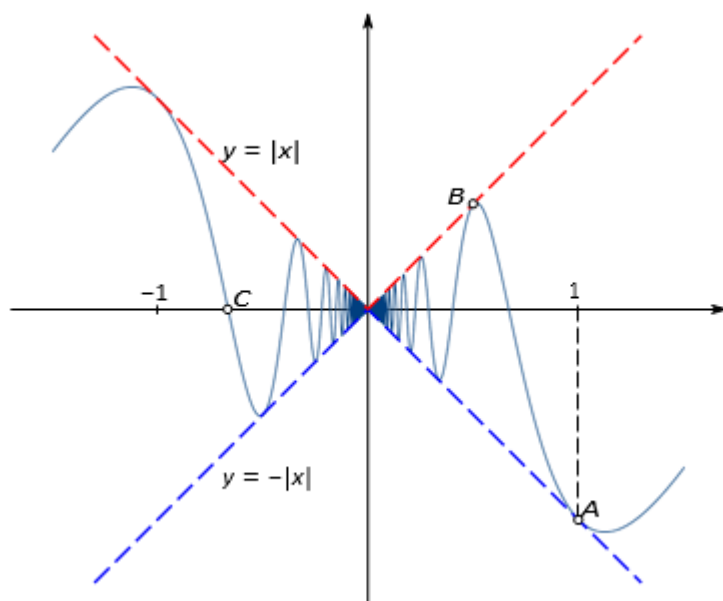


Figure 1.4: Graphs of $|x|$, $-|x|$ and $x \cos \frac{\pi}{x}$ for $-1.2 < x < 1.2$

(vii) Example: A Backward Cosine Sandwich. The sandwich Theorem says that if the function $g(x)$ is sandwich between two functions $f(x)$ and $h(x)$ and limit of the outside functions f and h exist and equal, then the limit of the inside function g exists and equals this common value. For example

$$-|x| \leq x \cos \frac{1}{x} \leq |x|$$

since the cosine is always between -1 and 1 . Since

$$\lim_{x \rightarrow 0} -|x| = \lim_{x \rightarrow 0} |x| = 0$$

the Sandwich Theorem tells us that,

$$\lim_{x \rightarrow a} x \cos \frac{1}{x} = 0$$

Note that the limit $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right)$ does **not** exist, for the same reason that the “backward sine” did not have a limit for $x \rightarrow 0$ (see example 7.3). multiplying with x changed that.

1.4: Definition of Continuity.

A function g is **continuous** at a if

$$\lim_{x \rightarrow a} g(x) = g(a) \quad (4)$$

A function is continuous if it is continuous at every a in its domain.

Note that when we say that a function is continuous on some interval it is understood that the domain of the function includes that interval. For example, the function $f(x) = \frac{1}{x^2}$ is continuous on the interval $1 < x < 5$ but is **not** continuous on the interval $-1 < x < 1$.

1.4.1: Polynomials are continuous.

For instance, let us show that $P(x) = x^2 + 3x$ is continuous at $x = 2$. To show that you have to prove that

$$\lim_{x \rightarrow 2} P(x) = P(2)$$

i.e.,

$$\lim_{x \rightarrow 2} x^2 + 3x = 3.2$$

You can do these two ways: using the definition with ε and δ (i.e., the hard way), or using the limit properties $(P_1) \dots (P_6)$ from 5 (just as good, and easier, even though it still takes a few lines to write it out-do both!).

1.4.2: Rational functions are continuous.

Let $R(x) = \frac{P(x)}{Q(x)}$ be a rational function, and let a be any number in the domain of R , i.e., any number for which $Q(a) \neq 0$. Then one has

$$\begin{aligned} \lim_{x \rightarrow a} R(x) &= \lim_{x \rightarrow a} \frac{P(x)}{Q(x)} \\ &= \frac{\lim_{x \rightarrow a} P(x)}{\lim_{x \rightarrow a} Q(x)} && \text{property } (P_6) \\ &= \frac{P(a)}{Q(a)} && P \text{ and } Q \text{ are continuous} \\ &= R(a) \end{aligned}$$

This shows that R is indeed continuous at a .

1.4.3: Discontinuous functions.

If $\lim_{x \rightarrow a} g(x)$ does not exist, then it certainly cannot be equal to $g(a)$, and

therefore any failed limit provides an example of a discontinuous function.

For instance, the sign function $g(x) = \text{sign}(x)$ from section 1.10 is not continuous at $x = 0$. Is the backward sine function $g(x) = \sin\left(\frac{1}{x}\right)$ also discontinuous at $x = 0$? No, it is not, for two reasons:

first, the limit $\lim_{x \rightarrow a} \sin\left(\frac{1}{x}\right)$ does not exist, and second, we haven't even defined the function $g(x)$ at $x = 0$, so even if the limit existed, we would have no value $g(0)$ to compare it with.

1.4.4: How to make functions discontinuous.

Here is a discontinuous function:

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 3, \\ 47 & \text{if } x = 3. \end{cases}$$

In other words, we take a continuous like $g(x) = x^2$, and change its value somewhere, e.g. at $x = 3$.

$$\lim_{x \rightarrow 3} f(x) = 9 \neq 47 = f(3)$$

The reason that the limit is 9 is that our new function

$f(x)$ coincides with our old continuous function $g(x)$ for all x except $x = 3$. Therefore the limit of $f(x)$ as $x \rightarrow 3$ is the same as the limit of $g(x)$ as $x \rightarrow 3$, and since g is continuous this is $g(3) = 9$.

1.4.5: Sandwich in a bow tie.

We return to the function from example.? Consider

$$f(x) = \begin{cases} x \cos \frac{1}{x} & \text{for } x \neq 0, \\ 0 & \text{for } x = 0 \end{cases}$$

Then f is continuous at $x = 0$ by the Sandwich Theorem (see Example ??).

If we change the definition of f by picking a different value at $x = 0$ the new function will not be continuous, since changing f at $x = 0$ does not change the limit $\lim_{x \rightarrow 0} f(x)$. Since this limit is zero $f(0) = 0$ is the only possible choice of $f(0)$ which makes f continuous at $x = 0$.

1.4.6: Substitution in Limits

Given function f and g one can consider their composition $h(x) = f(g(x))$. To compute the limit

$$\lim_{x \rightarrow a} f(g(x))$$

We write $u = g(x)$. So that we want to know

$$\lim_{x \rightarrow a} f(u) \quad \text{where } u = g(x)$$

Suppose that you can find the limits

$$L = \lim_{x \rightarrow a} g(x) \text{ and } \lim_{u \rightarrow L} f(u) = M.$$

Then it seems reasonable that as x approaches a , $u = g(x)$ will approach L , and $f(g(x))$ approaches M .

This is in fact a theorem:

1.4.7: Theorem: Composition of a Continuous Function and a Limit

if $\lim_{x \rightarrow a} g(x) = L$ and if the function f is continuous at $u = L$, then

$$\lim_{x \rightarrow a} f(g(x)) = \lim_{u \rightarrow L} f(u) = f(L)$$

Another way to write this is

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

(viii) **Example: Compute** $\lim_{x \rightarrow 3} \sqrt{x^3 - 3x^2 + 2}$. The given functions is the composition of two functions, namely

$$\sqrt{x^3 - 3x^2 + 2} = \sqrt{u}, \text{ with } u = x^3 - 3x^2 + 2,$$

or, in function notation, we want to find $\lim_{x \rightarrow 3} h(x)$ where

$$h(x) = f(g(x)), \text{ with } g(x) = x^3 - 3x^2 + 2 = \sqrt{2}$$

either way, we have

$$\lim_{x \rightarrow 3} x^3 - 3x^2 + 2 = 2 \text{ and } \lim_{x \rightarrow 3} \sqrt{u} = \sqrt{2}.$$

You get the first limit from the limit properties $(P_1) \dots (P_5)$. The second limit says that taking the square root is a continuous function, which it is. We have not proved that (yet), but this particular limit is the one from example 6.3. putting these two limits together we conclude that the limit is $\sqrt{2}$.

Normally, you write this whole argument as follows:

$$\lim_{x \rightarrow 3} \sqrt{x^3 - 3x^2 + 2} = \sqrt{\lim_{x \rightarrow 3} x^3 - 3x^2 + 2} = \sqrt{2}, \text{ where you must point out}$$

that $f(x) = \sqrt{x}$ is continuous to justify the first step.

Another possible way of writing this is

$$\lim_{x \rightarrow 3} \sqrt{x^3 - 3x^2 + 2} = \lim_{u \rightarrow 2} \sqrt{u} = \sqrt{2},$$

Where you must say that you have substituted

$$u = x^3 - 3x^2 + 2.$$

Exercises:

Find the following limits:

1. $(2x + 5)$

2. $(2x + 5)$

3. $(2x + 5)$

4. $(x + 3)^{2006}$

5. $(x + 3)^{2007}$

6. $(x + 3)$

7. $\frac{t^2+t-2}{t^2-1}$

8. $\frac{t^2+t-2}{t^2-1}$

9. $\frac{t^2+t-2}{t^2-1}$

10. $\frac{x^2+3}{x^2+4}$

11. $\frac{x^5+3}{x^2+4}$

13. $\frac{x^2+3}{x^5+4}$

14. $\frac{x^2+1}{x^5+4}$

15. $\frac{(2x+1)^4}{(3x^2+1)^2}$

16. $\frac{(2u+1)^4}{(3u^2+1)^2}$

17. $\frac{(2t+1)^4}{(3t^2+1)^2}$

18. what are the coordinates of the points labelled A, ...E in figure 1.3 (the graph of $y = \sin \frac{\pi}{x}$)
19. if the $\lim_{x \rightarrow a} f(x)$ exists then f is continuous at $x = a$. True or false?
20. Given two examples of functions for which $\lim_{x \rightarrow 0} f(x)$ does not exist
21. **Group Problem.**
If $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ both do not exist, then $\lim_{x \rightarrow 0} (f(x) + g(x))$ also does not exist. True or false?
22. **Group problem**
if $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ both do not exist, then $\lim_{x \rightarrow 0} \left(\frac{f(x)}{g(x)} \right)$ also does not exist. True or false?
23. **Group Problem**
In the text we proved that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Show that this implies that $\lim_{x \rightarrow \infty} x$ does not exist. Hint: suppose $\lim_{x \rightarrow \infty} x = L$ for some number L . Apply the limit properties to $\lim_{x \rightarrow \infty} x \cdot \frac{1}{x}$.
24. Evaluate $\lim_{x \rightarrow 9} \frac{\sqrt{x-3}}{x-9}$. Hint: Multiply top and bottom by $\sqrt{x+3}$.
25. Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{x}}{x-2}$.
26. Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{2}}}{x-2}$.

27. A function f is defined by

$$f(x) = \begin{cases} x^3 & \text{for } x < -1 \\ ax + b & \text{for } -1 \leq x < 1 \\ x^2 + 2 & \text{for } x \geq 1 \end{cases}$$

where a and b are constants. The function f is continuous. What are a and b ?

28. find a constant k such that the function

$$f(x) = \begin{cases} 3x + 2 & \text{for } x < 2 \\ x^2 + k & \text{for } x \geq 2 \end{cases}$$

is continuous. Hint: Compute the one-sided limits.

29. find constants a and c such that the function

$$f(x) = \begin{cases} x^3 + c & \text{for } x < 0 \\ ax + c^2 & \text{for } 0 \leq x < 1 \\ \arctan x & \text{for } x \geq 1 \end{cases}$$
 is continuous for all x

1.4.8: Two Limits in Trigonometry

In this section we'll derive a few limits involving the trigonometric functions. You can think of them as saying that for small angles θ one has

$$\sin \theta \approx \theta \text{ and } \cos \theta \approx 1 - \frac{1}{2}\theta^2$$

We will use the limits when we compute the derivatives of Sine, Cosine and Tangent.

1.4.9: Theorem $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$

Proof. The proof requires a few sandwiches and some geometry.

We begin by only considering positive angles, and in fact we will only consider angles $0 < \theta < \frac{\pi}{2}$.

Since the wedge OAC contains the triangle OAC its area must be larger. The area of the wedge is $\frac{1}{2}\theta$ and the area of the angle is $\frac{1}{2}\sin \theta$, so we find that

$$0 < \sin \theta < \theta \text{ for } 0 < \theta < \frac{\pi}{2} \quad (5)$$

The Sandwich Theorem implies that

$$\sin \theta = 0 \quad (6)$$

Moreover, we also have

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = 1 \quad (7)$$

Conclusion

In this session, we explored the concept of limits, a fundamental tool in calculus that helps us analyse the behaviour of functions as they approach specific values or infinity. By studying this session, you should now be able to define and use the keywords related to limits correctly. Additionally, you have gained an understanding of the significance of limits and functions in mathematics. You can now identify and analyse properties of limits, including continuity and one-sided limits, and confidently calculate the limit of a polynomial function using algebraic techniques.

Summary

1. A function is a correspondence from a first set, called the domain to a second set, called the range such that each element in the domain corresponds to exactly one element in the range.
2. A function is one -to -one if each value of the dependent variable in the range there corresponds exactly one value of the independent variable
3. If $f(x)$ becomes arbitrary close to a single number L as x approach c from either side, then $\lim_{x \rightarrow c} f(x) = L$ which is the limit of $f(x)$ as x approaches c is L .
4. If p is a polynomial function and c is any real number,
then $\lim_{x \rightarrow c} p(x) = p(c)$

Study Session 2: Derivatives

Introduction

This session covers the fundamental concepts of derivatives and their differentiability. We will explore the definition of derivatives, their notation, and various examples to deepen our understanding. Additionally, we will examine functions that are not differentiable at certain points and discuss the relationship between differentiability and continuity.

Learning Outcomes for Study Session 2:

When you have studied this session, you should be able to:

- 2.1. Define and use correctly all of the keywords printed in bold.
- 2.2. Explain the concept of Differentiation as the first principle.
- 2.3. Compute derivatives for expressions of the form x^n , where n is a constant exponent.
- 2.4. Perform differentiation on polynomial expressions.
- 2.5. Identify and apply standard derivatives to various functions.

2.1: Definition and Differentiability of Derivatives

Let f be a function which is defined on some interval (c, d) and let a be some number in this interval. The derivative of the function f at a is the value of the limit

$$f'(a) = \frac{f(x) - f(a)}{x - a}, \quad (8)$$

f is used to be differentiable at a if this limit exists.

f is called differentiable on the interval (c, d) if it is differentiable at every point a in (c, d) .

Other notations.

One can substitute $x = a + h$ in the limit (1) and let $h \rightarrow 0$ instead of $x \rightarrow a$. This gives the formula

$$f'(a) = \frac{f(a+h) - f(a)}{h}, \quad (9)$$

Often you will find this equation written with x instead of a and Δx instead of h , which makes it look like this:

$$f'(x) = \frac{f(x+\Delta x) - f(x)}{\Delta x}. \quad (10)$$

The interpretation is the same as in equation (6). The numerator $f(x + \Delta x) - f(x)$ represents the amount by which the function value of f changes if one increases its argument x by a (small) amount Δx . If you write $y = f(x)$ then we can call the increase in f .

$$\Delta y = f(x + \Delta x) - f(x),$$

so that the derivative $f'(x)$ is

$$f'(x) \frac{\Delta y}{\Delta x}$$

Gottfried Wilfried Von Leibniz, one of the inventors of Calculus, came up with the idea that one should write this limit as

$$\frac{dy}{dx} = \frac{\Delta y}{\Delta x},$$

The idea being that after letting Δx go to zero it didn't vanish, but instead became an infinity small quantity which Leibniz called " dx ". The result of increasing x by this infinity small quantity dx is that $y = f(x)$ increased by another infinity small quantity dy . The ratio of these two infinity small quantities is what we call derivative of $y = f(x)$.

There are "infinity small real numbers", and this makes Leibniz' notation difficult to justify. In the 20th century mathematicians have managed to create a consistent theory of "infinitesimals" which allows you to compute with " dx and dy " as Leibniz and his contemporaries would have done. This theory is called "non standard analysis". We won't mention it any further. Nonetheless, even though we won't use infinitely small numbers' notation is very small and we will use it. We next look at the direct computation of derivatives

2.1.2: Example – the derivative of $f(x) = x^2$ is $f'(x) = 2x$. We have done this computation before in (*). The result was

$$f'(x) = \frac{f(x+h)-f(x)}{h} = \frac{(x+h)^2-x^2}{h} = (2x + h) = 2x.$$

Leibniz would have written

$$\frac{dx^2}{dx} = 2x.$$

The derivative of $g(x) = x$ is $g'(x) = 1$.

Indeed, one has:

$$g'(x) = \frac{g(x+h)-g(x)}{h} = \frac{(x+h)-x}{h} = \frac{h}{h} = 1.$$

The derivative of any constant function is zero.

Let $k(x) = c$. Be a constant function. Then we have

$$k'(x) = \frac{k(x+h)-k(x)}{h} = \frac{c-c}{h} = 0 = 0.$$

Derivative of x^n for $n = 1, 2, 3, \dots$

To differentiate $f(x) = x^n$ one proceeds as follows:

$$f'(a) = \frac{f(x)-f(a)}{x-a} = \frac{x^n-a^n}{x-a}.$$

We need to simplify the fraction $(x^n - a^n)/(x - a)$. For $n = 2$ we have

$$\frac{x^2-a^2}{x-a} = x + a$$

For $n = 1, 2, 3, \dots$ the geometric sum formula tells us that

$$\frac{x^n-a^n}{x-a} = x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + xa^{n-2} + a^{n-1}. \quad (11)$$

If you don't remember the geometric sum formula, then you could also just verify (4) by carefully multiplying both sides *with* $x - a$.

For instance, $n = 3$ you would get:

$$x \times (x^2 + xa + a^2) = x^3 + ax^2 + a^2x \quad \frac{-a \times (x^2 + xa + a^2)}{(x-a) \times (x^2 + xa + a^2)} = \frac{-ax^2 - a^2x - a^3}{x^3 - a^3}$$

With formula (4) in hand we can now easily find the derivative of x^n :

$$\begin{aligned} f'(a) &= \frac{x^n - a^n}{x - a} \\ &= \{x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + xa^{n-2} + a^{n-1}\} \\ &= a^{n-1} + a^{n-2}a + a^{n-3}a^2 + \dots + aa^{n-2} + a^{n-1}. \end{aligned}$$

Here there are n terms and they all are equal to a^{n-1} , so that final result is

$$f'(a) = na^{n-1}$$

One could also write this as $f'(x) = nx^{n-1}$, in Leibniz notation.

$$\frac{dx^n}{dx} = na^{n-1}.$$

This formula turns out to be true in general, but here we have only proved it for the case in which n is a positive integer. The next theorem shows how differentiable implies continuity.

2.1.6: Theorem. If a function f is differentiable at some a in its domain, then f is also continuous at a .

Proof. We are given that

$$\frac{f(x) - f(a)}{x - a}$$

exists, and we must show that

$$f(x) = f(a)$$

This follows from the following computation

$$\begin{aligned} f(x) &= (f(x) - f(a) + f(a)) \quad (\text{algebra}) \\ &= \frac{f(x) - f(a)}{x - a} \cdot (x - a) + f(a) \end{aligned}$$

$$= \left\{ \frac{f(x) - f(a)}{x - a} \right\} (x - a) - f(a)$$

$$(\text{Limit properties}) = f'(a) \cdot 0 + f(a)$$

2.2: Some non-differentiable functions

2.8.1 A graph with a corner. $f(x) = |x| = \begin{cases} x & \text{for } x \geq 0, \\ -x & \text{for } x < 0. \end{cases}$

This function is continuous at all x , but it is not differentiable at $x = 0$.

$$f'(0) = \frac{|x| - |0|}{x - 0} = \frac{|x|}{x} = \sin(x)$$

If you look at the graph of $f(x) = |x|$ Then you see what is wrong: the graph has a corner at the origin and it is not clear which line, if any, deserves to be called the tangent to the graph at the origin.

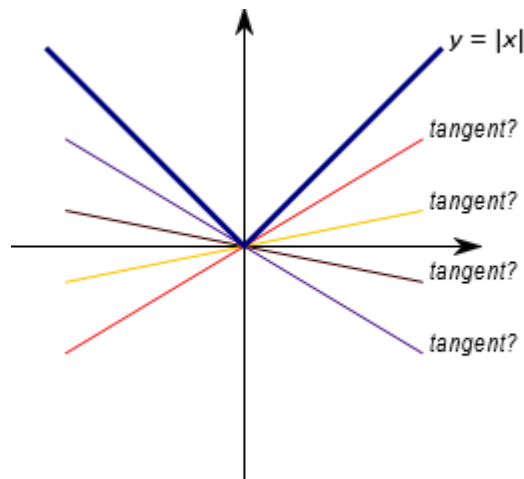


Figure 2.1. The graph of $y = |x|$ has no tangent at the origin.

2.8.2 A graph with a cusp. Another example of a function without a derivative at $x = 0$ is

$$f(x) = \sqrt{|x|}.$$

When you try to compute the derivative you get this limit.

$$f'(0) \frac{\sqrt{|x|}}{x} = ?$$

Then the limit from the right is

$$\frac{\sqrt{|x|}}{x} = \frac{1}{\sqrt{x}}.$$

Which does not exist (*it is* $+\infty$). Likewise, the limit from the left also does not exist (*it is* $-\infty$). Nonetheless, a drawing for the graph of f suggests an obvious tangent to the graph at $x = 0$, namely, the y - *axis*. That observation does not give us a derivative, because the y - *axis* is vertical and hence has no slope.

A graph with absolutely no tangents, anywhere. The previous two examples were about functions which did not have a derivative at $x = 0$. In both examples the point $x = 0$ was the only point where the function failed to have a derivative. It is easy to give examples of functions which are not differentiable at more than one value of x , but here I would like to show you a function f which doesn't have a derivative anywhere in its domain. This function exhibits a phenomenon known as a "discontinuity of the derivative." The absence of a derivative at any point within its domain is a striking characteristic of this function, challenging our intuition about tangent lines and slopes.

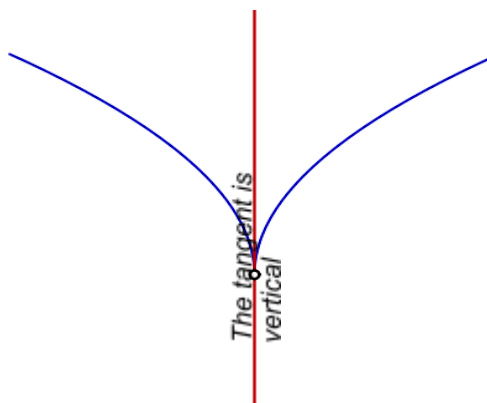


Figure 2.2. Tangent to the graph of $y = |x|^{\frac{1}{2}}$ at the origin

Exercises:

1. Compute the derivative of the following functions

$$(i) f(x) = x^2 - 2x. \quad (ii) g(x) = \frac{1}{x} \quad (iii) k(x) = x^3 - 17x \quad (iv)$$

$$u(x) = \frac{2}{1+x}$$

$$(v) v(x) = \sqrt{x} \quad (vi) w(x) = \frac{1}{\sqrt{x}}.$$

2. Which of the following functions is differentiable at $x = 0$?

$$(i) f(x) = x|x| = x\sqrt{|x|}, \quad (ii) h(x) = x + |x|, \quad (iii) k(x) = x^2 \sin \frac{\pi}{x},$$

$$(iv) l(x) = x \sin \frac{\pi}{x}.$$

These formulas do not define

k and l at $x = 0$. We define $k(0) = l(0) = 0$.

3. For which value(s) is the function defined by

$$f(x) = \begin{cases} ax+b & \text{for } x < 0 \\ x-x^2 & \text{for } x \geq 0 \end{cases}$$

differentiable at $x = 0$? Sketch the graph of the function f for the values a and b you found.

4. For which value(s) is the function defined by

$$f(x) = \begin{cases} ax^2 + b & \text{for } x < 1 \\ x - x^2 & \text{for } x \geq 1 \end{cases}$$

differentiable at $x = 0$? Sketch the graph of the function f for the values a and b you found.

5. For which value(s) is the function defined by

$$f(x) = \begin{cases} ax^2 & \text{for } x < 2 \\ x + b & \text{for } x \geq 2 \end{cases}$$

differentiable at $x = 0$? Sketch the graph of the function f for the values a and b you found.

6. Group problem. *True or false*: If a function f is continuous at some $x = a$ then it must also be differentiable at $x = a$?
7. Group problem. *True or false*: If a function f is differentiable at some $x = a$ then it must also be continuous at $x = a$?

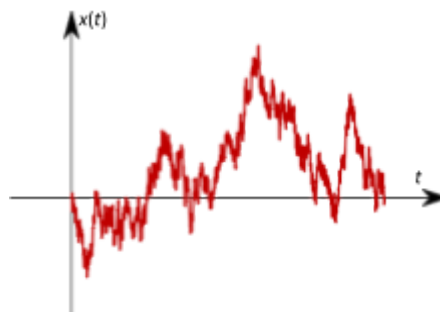


Figure 2.3. A Brownian motion.

Note how the graph doesn't seem to have a tangent anywhere at all

Conclusion

In this session, we explored the definition and differentiability of derivatives. The derivative of a function f at a point a is given by the limit $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$, and a function is differentiable at a if this limit exists. We computed derivatives for specific functions and discussed examples of non-differentiable functions, encountering cases with corners, cusps, and functions lacking derivatives everywhere in their domains. The study of derivatives is fundamental in understanding the behaviour of functions and their rates of change, with applications in various fields such as physics, economics, and engineering.

Summary

1. Differentiation is the action of computing a derivative. The derivative of a function $f(x)$ of a variable x is a measure of the rate at which the changes with respect to the change of the variable.
2. Differentiation of polynomial means differentiating each term in turn

Study Session 3: Techniques of Differentiation

Introduction

Continuing to compute derivatives using the definition can be a tedious task as it requires comparing new limits and hoping for a clever technique to evaluate them. However, there is some good news – such an approach is not always necessary. It turns out that if you know a few basic derivatives (such as $dx^n/dx = nx^{n-1}$) then you can find derivatives of arbitrarily complicated functions by breaking them into smaller pieces. In this section, we will look at rules which tell you how to differentiate a function which is either the sum, difference, product or quotient of two other functions.

Constant rule: $c' = 0 \quad \frac{dc}{dx} = 0$

Sum rule: $(u \pm v)' = u' \pm v' \quad \frac{du \pm v}{dx} = \frac{du}{dx} \pm \frac{dv}{dx}$

Product Rule: $(u \pm v)' = u' \cdot v + u \cdot v' \quad \frac{d(uv)}{dx} = \frac{du}{dx} v \pm u \frac{dv}{dx}$

Quotient rule: $\left(\frac{u}{v}\right)' = \frac{u' \cdot v - u \cdot v'}{v^2} \quad \frac{d\frac{u}{v}}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Learning Outcomes for Study Session 3:

When you have studied this session, you should be able to:

- 3.1. Define and use correctly all of the keywords printed in bold.
- 3.2. Apply the product rule to differentiate products of functions.
- 3.3. Apply the quotient rule to differentiate quotients of two functions.
- 3.4. Understand and apply the concept of composite functions.
- 3.5. Differentiate implicit functions.

3.1: Sum, Product and Quotient rules.

In the following c and n are constants, u , and v are functions of x , and $'$ denotes differentiation. The Differentiation Rules in function notation, and Leibniz notation.

3.1.1: Proof of the Sum Rule.

Suppose that $f(x) = u(x) + v(x)$ for all x where u and v are differentiable. Then

$$\begin{aligned} f'(a) &= \frac{f(x)-f(a)}{x-a} \quad (\text{definition of } f') \\ &= (u(x) + v(x)) - (u(a) + v(a)) \quad (\text{use } f = u + v) \\ &= \left(\frac{u(x)-u(a)}{x-a} + \frac{v(x)-v(a)}{x-a} \right) \quad (\text{algebra}) \\ &= \frac{u(x)-u(a)}{x-a} + \frac{v(x)-v(a)}{x-a} \quad (\text{limit property}) \\ &= u'(a) + v'(a) \quad (\text{definition of } u', v') \end{aligned}$$

3.1.2: Proof of the Product Rule.

Let $f(x) = u(x)v(x)$. To find the derivative, we must express the change of f in terms of the changes of u and v .

$$\begin{aligned} f(x) - f(a) &= u(x)v(x) - u(a)v(a) \\ &= u(x)v(x) - u(x)v(a) + u(x)v(a) - u(a)v(a) \\ &= u(x)(v(x) - v(a)) + (u(x) - u(a))v(a) \end{aligned}$$

Now divide by $x - a$ and let $x \rightarrow a$:

$$\frac{f(x)-f(a)}{x-a} = u(x) \frac{v(x)-v(a)}{x-a} + \frac{u(x)-u(a)}{x-a} v(a)$$

(Use the limit properties)

$$= (u(x)) \left(\frac{u(x)-v(a)}{x-a} \right) + \left(\frac{u(x)-u(a)}{x-a} \right) v(a)$$

$$= u(a)v'(a) + u'(a)v(a),$$

as claimed. In this last step we have used that

$$\frac{u(x)-u(a)}{x-a} = u'(a) \text{ and } \frac{u(x)-v(a)}{x-a} = v'(a)$$

and also, that

$$u(x) = u(a)$$

This last limit follows from the fact that u is continuous, which in turn follows from the fact that u is differentiable.

3.1.3: Proof of the Quotient Rule.

We can break the proof into two parts. First, we do the special case where $f(x) = 1/v(x)$, and then we use the product rule to differentiate

$$f(x) = \frac{u(x)}{v(x)} = u(x) \cdot \frac{1}{v(x)}.$$

So let $f(x) = 1/v(x)$. We can express the change in f in terms of the change in v

$$f(x) - f(a) = \frac{1}{v(x)} - \frac{1}{v(a)} = \frac{v(x)-v(a)}{v(x)v(a)}.$$

Dividing by $x - a$ we get

$$\frac{f(x)-f(a)}{x-a} = \frac{1}{v(x)v(a)} \frac{v(x)-v(a)}{x-a}$$

Now we want to take the limit $x \rightarrow a$. We are given the v is differentiable, so it must also be continuous and hence

$$v(x) = v(a).$$

Therefore, we find

$$\frac{f(x)-f(a)}{x-a} = \frac{1}{v(x)v(a)} \frac{v(x)-v(a)}{x-a} = \lim_{x \rightarrow a} \frac{v'(a)}{v(a)^2}$$

This completes the first step of the proof. In the second step we use the product rule to differentiate $f = u/v$

$$f'\left(\frac{u}{v}\right) = \left(u, \frac{1}{v}\right)' = u' \cdot \frac{1}{v} + u\left(\frac{1}{v}\right)' = \frac{u'}{v} - u \frac{v'}{v^2} = \frac{u'v - uv'}{v^2}$$

3.1.4: A shorter, but not quite perfect derivation of the Quotient Rule.

The quotient Rule can be derived from the Product Rule as follows:

if $w = u/v$ then

$$w \cdot v = u \tag{12}$$

By the product rule we have

$$w' \cdot v + w \cdot v' = u'$$

so that

$$w' = \frac{u' - w \cdot v'}{v} = \frac{u' - (u/v) \cdot v'}{v} = \frac{u'v - u \cdot v'}{v^2}$$

Unlike the proof in the equation above, this argument does not prove that w is differentiable if u and v are. It only says that **if the derivative exists**, then it must be what the Quotient Rule says. Note that the rule $(cu)' = cu'$ follows from the constant rule and the product rule.

If u and v are quantities which depends on x , and if increasing x by Δx causes u and v to change by Δu and Δv , then the product rule of u and v will change by

$$\Delta(uv) = (u + \Delta u)(v + \Delta v) - uv = u\Delta v + \Delta u\Delta v + v\Delta u \tag{13}$$

If u and v are differentiable functions of x , then the changes Δu and Δv will be of the same order of magnitude as Δx , and thus expects $\Delta u \Delta v$ to be much smaller. One therefore ignores the last term in (6), and thus arrives at

$$\Delta(uv) = u\Delta v + v\Delta u.$$

Leibniz would now divide by Δx and replace Δ 's by d 's to get the product rule:

$$\Delta(uv) = u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x}.$$

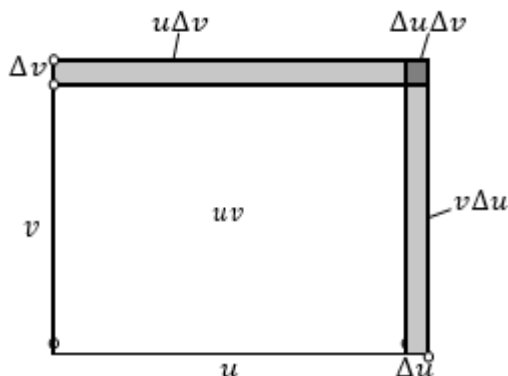


Figure 3.1. The Product Rule.

How much does the area of a rectangle change if its sides u and v are increased by Δu and Δv ? Most of the increase is accounted for by the two thin rectangles whose areas are $u\Delta v$ and $v\Delta u$. So the increase in area is approximately $u\Delta v + v\Delta u$, which explains why the product rule says $(uv)' = uv' + vu'$.

3.2: Differentiating powers of functions

3.2.1: Product rule with more than one factor.

If a function is the product of n functions i.e.

$$f(x) = u_1(x) \times u_2(x) \dots \times u_n(x),$$

Then you can differentiate it by applying the product rule $n - 1$ times (there are n factors, so there are $n - 1$ multiplications.)

After the first step you would get

$$f' = u_1'(u_2 \dots u_n) + u_1(u_2' \dots u_n).$$

In the second step up apply the product rule to $(u_2 u_3 \dots u_n)'$. This yields

$$\begin{aligned} f' &= u_1' u_2 \dots u_n + u_1 [u_2' u_3 \dots u_n + u_2 (u_3' \dots u_n)'] \\ &= u_1' u_2 \dots u_n + u_1 u_2' u_3 \dots u_n + u_1 u_2 (u_3' \dots u_n)'. \end{aligned}$$

Continuing this way, one finds after $n - 1$ applications of the product rule that

$$(u_1 \dots u_n)' = u_1' u_2 \dots u_n + u_1 u_2' u_3 \dots u_n + \dots + u_1 u_2 u_3 \dots u_n' \quad (14)$$

3.2.2: The Power Rule.

If all n factors in the previous paragraph are the same, so the function f is the n^{th} power of some other function.

$$f(x) = (u(x))^n,$$

then all terms in the right-hand side of the above function are the same, and, since are n of them, one gets

$$f' = n u^{n-1}(x) u'(x)$$

or, in Leibniz' notation,

$$\frac{du^n}{dx} = nu^{n-1} \frac{du}{dx}. \quad (15)$$

3.2.3: The Power Rule in Negative Integer Exponents.

We have just moved the power rule assuming n is a positive integer. The rule actually holds for all exponents n , but the proof is harder. Here we prove the Power Rule for negative exponents using the Quotient Rule. Suppose $n = -m$ where m is a positive integer. Then the Quotient Rule tells us that

$$(u^n)' = (u^{-m})' = \left(\frac{1}{u^m}\right)' \cdot Q \cdot R = \frac{(u^m)'}{(u^m)^2}$$

Since m is a positive integer, we can use (8), so $(u^m)' = mu^{m-1}$ and hence

$$(u^n)' = \frac{mu^{m-1} \cdot u'}{u^{2m}} = -mu^{m-1} \cdot u' = nu^{n-1} u'.$$

3.2.4: The Power Rule for Rational Exponents.

So far we proved that the power law holds if the exponent n is an integer. We will now see how you can show that the power law holds even if the exponent n is any fraction, $n = p/q$. The following derivation contains the trick called **implicit differentiation**. So let $n = p/q$ where p and q are integers and consider the function

$$w(x) = u(x)^{p/q}.$$

Assuming that both u and w are differentiable functions, we will show that

$$w'(x) = \frac{p}{q} u(x)^{\frac{p}{q}-1} u'(x) \quad (16)$$

Raising both sides to the q^{th} power gives

$$w(x)^q = u(x)^p.$$

Here the exponents p and q are integers, so we may apply the Power Rule to both sides. We get

$$qw^{q-1} \cdot w' = pu^{p-1} \cdot u'$$

Dividing both sides by qw^{q-1} and substituting $u^{p/q}$ for w gives

$$w' = \frac{pu^{p-1} \cdot u'}{qw^{q-1}} = \frac{pu^{p-1} \cdot u'}{qu^{q-1}} = \frac{pu^{p-1} \cdot u'}{qu^{p-(p/q)}} = \frac{p}{q} \cdot u^{(p/q)-1} \cdot u'.$$

Which is the Power Rule for $n = p/q$.

3.2.5: Derivative of x^n for integer n .

If you choose the function $u(x)$ in the Power Rule to be $u(x) = x$, then $u'(x) = 1$, and hence the derivative of $f(x) = u(x)^n = x^n$ is

$$f'(x) = nu(x)^{n-1} u'(x) = nx^{n-1} \cdot 1 = nx^{n-1}.$$

Example – differentiate a polynomial. Using the Differentiation Rules you can easily differentiate any polynomial and hence any rational function. For example, using the Sum Rule, the Power Rule with $u(x) = x$, the rule $(cu)' = cu'$, the derivative of the polynomial

$$f(x) = 2x^4 - x^3 + 7$$

is

$$f'(x) = 8x^3 - 3x^2$$

Example - differentiate a rational function. By the Quotient Rule the derivative of the function.

$$g(x) = \frac{2x^4 - x^3 + 7}{1+x^2}$$

is

$$g'(x) = \frac{(8x^3 - 3x^2)(1+x^2) - (2x^4 - x^3 + 7)2x}{(1+x^2)^2}$$

$$\frac{6x^5 - x^4 + 8x^3 - 3x^2 - 14x}{(1+x^2)^2}$$

Derivative of the square root. The derivative of $f(x) = \sqrt{x} = x^{1/2}$ is

$$f'(x) = \frac{1}{2}x^{1/2-1} = \frac{1}{2}x^{-1/2} = \frac{1}{2x^{1/2}} = \frac{1}{2\sqrt{x}}$$

when we used the power rule with $n = 1/2$ and $u(x) = x$.

Exercise

- Let $f(x) = (x^2 + 1)(x^3 + 3)$ Find $f'(x)$ in two ways:
 - By multiplying and then differentiating
 - By using the product rule. Are your answers the same?
- Let $f(x) = (1 + x^2)^4$. Find $f'(x)$ in two ways, first by expanding to get an expression for $f(x)$ as a polynomial in x and then differentiating and then by using the power rule. Are the answers the same?
- Prove the statement $(cu)' = c(cu)'$ follows from the product rule. Compute the derivatives of the following functions.

$$(a) \quad f(x) = x + 1 + (x + 1)^2; \quad (b) \quad f(x) = \frac{x-2}{x^4+1}; \quad (c) \quad f(x) = \left(\frac{1}{1+x}\right)^{-1};$$

$$(d) \quad f(x) = \sqrt{1-x^2}; \quad (e) \quad f(x) = \frac{ax+b}{cx+d}; \quad (f) \quad f(x) = \frac{1}{(1+x^2)^2}; \quad (g)$$

$$f(x) = \frac{x}{1+\sqrt{x}}; \quad (h) \quad f(x) = \sqrt{\frac{1-x}{1+x}}; \quad (i) \quad f(x) = \sqrt[3]{x + \sqrt{x}}; \quad (j) \quad \varphi(t) = \frac{t}{1+\sqrt{t}};$$

$$(k) \quad g(s) = \sqrt{\frac{1-s}{1+s}}; \quad (l) \quad h(p) = \sqrt[3]{p} + \sqrt{p}$$

3.3: Higher Derivatives

The derivative $f'(a)$ of some function f exists for all a in the domain of f , then we have a new function; namely, for each number in the domain of f we compute the derivative of f at that number. This function is called the **derivative function** of f , and it is denoted by f' . Now that we have agreed on the derivative of a function, we can repeat the process and try to differentiate the derivative. The result, if it exists, is called the **second derivative of f** . It is denoted f'' . The derivative of the second derivative is called the third derivative, written f''' , and so on. The n^{th} derivative of f is denoted $f^{(n)}$. Thus

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(3)} = f''', \dots$$

Leibniz notation for the n^{th} derivative of $y = f(x)$ is

$$\frac{d^n y}{dx^n} = f^{(n)}(x).$$

Example. If $f(x) = x^2 - 2x + 3$ then

$$f(x) = x^2 - 2x + 3 \quad f'(x) = 2x - 2$$

$$f''(x) = 2$$

$$f^{(3)}(x) = 0$$

$$f^{(4)}(x) = 0$$

All further derivatives of f are zero.

Operator notation. A common variation on Leibniz notation for derivatives is the so-called operator notation, as in

$$\frac{d(x^3 - x)}{dx} = \frac{d}{dx}(x^3 - x) = 3x^2 - 1.$$

For higher derivatives one can write

$$\frac{d^2 y}{dx^2} = \left(\frac{dy}{dx} \right)^2 y$$

Be careful to distinguish the second derivative from the square of the first derivative. Usually

$$\frac{d^2 y}{dx^2} \neq \left(\frac{dy}{dx} \right)^2 !!!!$$

Exercise:

- The equation $\frac{2x}{x^2 - 1} = \frac{1}{x+1} + \frac{1}{x-1}$ holds for all values of x (except $x = \pm 1$), so you should get the same answer if you differentiate both sides. Compute the third derivative of $f(x) = 2x/(x^2 - 1)$ by using either the left or right-hand side.

2. Compute the first, second and third derivatives of the following functions. (a) $f(x) = (x + 1)^4$; (b) $g(x) = (x^2 + 1)^4$; © $h(x) = \sqrt{x - 2}$; (d) $k(x) = \sqrt[3]{x - \frac{1}{x}}$.
3. Find the derivatives of the 10th order of the functions. (a) $f(x) = x^{12} + x^8$; (b) $g(x) = \frac{1}{x}$; © $h(x) = \frac{12}{1-x}$; (d) $k(x) = \frac{x^2}{1-x}$.
4. Find $f'(x)$, $f''(x)$ and $f^{(3)}(x)$ if $f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6}{720}$.

3.4: Differentiating Trigonometric functions

The trigonometric functions Sine, Cosine and Tangent are differentiable, and their derivatives are given by the following formulas.

$$\frac{d \sin x}{dx} = \cos x, \quad \frac{d \cos x}{dx} = -\sin x, \quad \frac{d \tan x}{dx} = \frac{1}{\cos^2 x} \quad (17)$$

Note the minus sign in the derivative of the cosine!

Proof. By definition one has

$$\sin'(x) = \frac{\sin(x+h) - \sin(x)}{h}.$$

To simplify the numerator, we use the trigonometric addition formula

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

with α and $\beta = h$, which results in

$$\frac{\sin(x+h) - \sin(x)}{h} = \frac{\sin(x) \cos(h) + \cos(x) \sin(h) - \sin(x)}{h}$$

$$= \cos \cos(x) \frac{\sin \sin(h)}{h} + \sin(x) \frac{\cos \cos(h) - 1}{h}$$

Hence the formulas

$$\frac{\sin(h)}{h} = 1 \text{ and } \frac{\cos \cos(h) - 1}{h} = 0$$

Recall that

$$\sin'(x) = \cos \cos(x) \frac{\sin \sin(h)}{h} + \sin(x) \frac{\cos \cos(h) - 1}{h}$$

$$= \cos \cos(x) \cdot 1 + \sin \sin(x) \cdot 0$$

$$= \cos \cos(x).$$

A similar computation leads to the stated derivatives of $\cos x$. To find the derivative of $\tan x$ we apply the quotient rule to

$$\tan x = \frac{\sin x}{\cos x} = \frac{f(x)}{g(x)}.$$

We get

$$\tan'(x) = \frac{\cos(x) \sin'(x) - \sin(x) \cos'(x)}{\cos^2(x)} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)}$$

Exercise:

1. Find the derivatives of the following functions.

i). $f(x) = \sin \sin(x) + \cos(x)$; (ii) $f(x) = 2 \sin \sin(x) - 2 \cos(x)$; (iii)

$f(x) = 3 \sin \sin(x) + 2 \cos(x)$; $f(x) = x \sin(x) + \cos(x)$; (iv)

$f(x) = x \cos(x) - \sin \sin x$; (v) $f(x) = \frac{\sin \sin x}{x}$; (vi) $f(x) = \cos^2(x)$; (vii)

$f(x) = \sqrt{1 - \sin^2 x}$; (viii) $f(x) = \sqrt{\frac{1 - \sin \sin x}{1 + \sin \sin x}}$; (ix) $\cot \cot(x) = \frac{\cos \cos x}{\sin \sin x}$;

2. Find a and b so that the function $f(x) = \begin{cases} \cos \cos x & \text{for } x \leq \frac{\pi}{4} \\ a + bx & \text{for } x > \frac{\pi}{4} \end{cases}$ is differentiable

at $x = \pi/4$?

3. Find a and b so that the function $f(x) = \begin{cases} \tan x & \text{for } x < \frac{\pi}{6} \\ a+bx & \text{for } x \geq \frac{\pi}{6} \end{cases}$ is differentiable at $x = \pi/6$?

3.5: The Chain Rule

Composition of functions. Given two functions f and g , one can define a new function called the *composition of f and g* . The notation for the composition is $f \circ g$, and it is defined by the formula.

$$f \circ g(x) = f(g(x)).$$

The domain of the composition is the set of all numbers x for which this formula gives you something well-defined.

For instance, if $f(x) = x^2 + x$ and $g(x) = 2x + 1$ then

$$f \circ g(x) = f(2x + 1) = (2x + 1)^2 + (2x + 1)$$

$$\text{and } g \circ f(x) = g(x^2 + x) = 2(x^2 + x) + 1$$

Note that, $f \circ g$ and $g \circ f$ are not the same function in this example (they hardly ever are the same). If you think of function as expressing dependency of one quantity on another, then the composition of functions arises as follows. If a quantity z is a function of another quantity if y itself depends on x , then z depends on x via y .

To get $f \circ g$ from the previous example, we could say $z = f(y)$ and $y = g(x)$, so that

$$z = f(y) = y^2 + y \text{ and } y = 2x + 1$$

Give x one can compute y , and from y one can then compute z . The result will be

$$z = y^2 + y = (2x + 1)^2 + (2x + 1),$$

In another notation,

$$z = f(y) = f(g(x)) = f \circ g(x).$$

One says that, the composition of f and g is the result of substituting g in f .

3.5.1: Theorem (Chain Rule)

if f and g are differentiable, so is the composition $f \circ g$.

The derivative of $f \circ g$ is given by

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

The chain rule tells you how to find the derivative of the composition $f \circ g$ of two functions f and g provided you now know how to differentiate the two functions f and g .

When written in Leibniz' notation the chain rule looks particularly easy. Suppose that $y = g(x)$ and $z = f(y)$, then $z = f \circ g(x)$, and the derivative of z with respect to x is the derivative of the function $f \circ g$. The derivative of z with respect to y is the derivative of the function f , and the derivative of y with respect to x is the derivative of the function g . In short,

$$\frac{dz}{dx} = (f \circ g)'(x), \quad \frac{dz}{dy} = f'(y) \quad \text{and} \quad \frac{dy}{dx} = g'(x)$$

So that, the chain rule says

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} \tag{18}$$

First Proof of The Chain Rule (Using Notation).

We first consider difference quotients instead of derivatives, i.e., using the same notation as above, we consider the effect of an increase of x by an amount Δx on the quantity z .

$$\Delta y = g(x + \Delta x) - g(x),$$

and $z = f(y)$ will increase by

$$\Delta z = f(y + \Delta y) - f(y),$$

The ratio of the increase in $z = f(g(x))$ to the increase in x is

$$\frac{\Delta z}{\Delta x} \rightarrow \frac{dz}{dx}, \quad \frac{\Delta z}{\Delta x} \rightarrow \frac{dz}{dy}, \quad \frac{\Delta y}{\Delta x} \rightarrow \frac{dy}{dx}$$

which immediately leads to Leibniz' form of the quotient rule.

Proof of the Chain Rule: We verify the formula in Theorem of chain rule above with value $x = a$, i. e we will show that

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

By definition the left-hand side is

$$(f \circ g)'(a) = \frac{(f \circ g)(x) - (f \circ g)(a)}{x - a} = \frac{f(g(x)) - f(g(a))}{x - a}$$

The two derivatives on the right-hand side are given by

$$g'(a) = \frac{g(x) - g(a)}{x - a}$$

and

$$f'(g(a)) = \frac{f(y) - f(g(a))}{y - a}$$

since g is a differentiable function, it must also be a continuous function, and hence $g(x) = g(a)$. So we can substitute $y = g(x)$ in the limit defining $f'(g(a))$.

Putting all this together yields

$$\begin{aligned}(f \circ g)'(a) &= \frac{f(g(x)) - f(g(a))}{x - a} \\&= \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} \\&= \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} \\&= f'(g(a)) \cdot g'(a)\end{aligned}$$

which is what we were supposed to prove – the proof seems complete.

There is one flaw in this proof, namely, we have divided $g(x) - g(a)$, which is not allowed when $g(x) - g(a) = 0$. This flaw can be fixed but we will not go into the details here.

Example.

If a function

$$z = f(y) = y^2 + y \text{ and } y = g(x) = 2x + 1$$

from the beginning of this section. The composition of these two functions is

$$z = f(g(x)) = (2x + 1)^2 + (2x + 1) = 4x^2 + 6x + 2$$

we compute the derivative of this composed function, i.e the derivative of z with respect to x in two ways. First, you simply differentiate the last formula we have:

$$\frac{dz}{dx} = \frac{d(4x^2 + 6x + 2)}{dx} = 8x + 6 \quad (19)$$

The other approach is to use the chain rule:

$$\frac{dz}{dx} = \frac{d(y^2 + y)}{dy} = 2y + 1,$$

and

$$\frac{dy}{dx} = \frac{d(2x+1)}{dx} = 2$$

Hence, by the chain rule one has

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = (2y + 1) \cdot 2 = 4y + 2 \quad (20)$$

The two answers (19) and (20) should be the same. Once you remember that $y = 2x + 1$ you see that this is indeed true.

$$y = 2x + 1 \Rightarrow 4y + 2 = 4(2x + 1) + 2 = 8x + 6$$

The two computations of dz/dx therefore lead to the same answer. In this example, there is no clear advantage in using chain rule. The chain rule becomes useful when the functions f and g becomes more complicated.

$$h(y) = \begin{cases} \frac{f(y)-f(g(a))}{y-g(a)} & u \neq a \\ f'(g(a)) & y=a \end{cases}$$

Example where you really need the Chain Rule. We know what the derivative of $\sin s$ with respect to x is, but none of the rules we have found so far tell us how to differentiate $f(x) = \sin(2x)$.

The function $f(x) = \sin \sin 2x$ is the composition of two simpler functions, namely

$$f(x) = g(h(x)) \text{ where } g(u) = \sin \sin u \text{ and } h(x) = 2x.$$

We know how to differentiate each of the two functions g and h :

$$g'(u) = \cos \cos u, \quad h'(x) = 2.$$

Therefore, the chain rule implies that

$$f'(x) = g'(h(x))h'(x) = \cos \cos (2x) \cdot 2 = 2\cos 2x.$$

Leibniz would have decomposed the relation $y = \sin \sin 2x$ between y and x as

$$y = \sin \sin u, \quad u = 2x$$

and then computed the derivative of $\sin \sin 2x$ with respect to x as follows

$$\frac{d \sin \sin x}{dx} = \frac{d \sin \sin u}{dx} = \frac{d \sin \sin u}{du} \cdot \frac{du}{dx} = \cos \cos u \cdot 2 = 2 \cos \cos 2x.$$

3.5.2: The Power Rule and the Chain Rule.

The Power Rule, says that for any function f and any rational number n one has

$$\frac{d}{dx} (f(x)^n) = n f(x)^{n-1} f'(x),$$

is a special case of the Chain Rule, for one regard $y = f(x)^n$ as the composition of two functions

$$\frac{du^n}{dx} = \frac{du^n}{du} \cdot \frac{du}{dx} = n u^{n-1} \frac{du}{dx}.$$

Setting $u = f(x)$ and $\frac{du}{dx} = f'(x)$ then gives you the Power Rule.

Example: The Volume of an inflating balloon.

Consider the “real world example” with radius $r = f(t)$. the volume of this Balloon is

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi f(t)^3.$$

We can regard this as the composition of two functions,

$$V = g(r) = \frac{4}{3}\pi r^3 \text{ and } r = f(t).$$

According to the chain rule the rate of change of the volume with the time is now

$$\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt}$$

i.e., it is the product of the rate of change of the volume with the radius of the balloon and the rate of change of the balloons with time. From

$$\frac{dV}{dr} = \frac{\frac{d}{dr}\left(\frac{4}{3}\pi r^3\right)}{dr} = 4\pi r^2$$

we see that,

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

For instance, if the radius of the balloon is growing at 0.5inch/sec, and if its radius is $r = 3.0 \text{ inch}$, then the volume is growing at the rate of

$$\frac{dV}{dt} = 4\pi(3.0\text{inch})^2 \times \frac{0.5\text{inch}}{\text{sec}} \approx \frac{57\text{inch}^3}{\text{sec}}.$$

A more complicated example. Suppose you needed to find the derivative of

$$y = h(x) = \frac{\sqrt{x+1}}{(\sqrt{x+1}+1)^2}$$

we can write this function as a composition of two simpler functions, namely,

$$y = f(u), u = g(x),$$

with

$$f(u) = \frac{u}{(u+1)^2} \text{ and } g(x) = \sqrt{x+1}.$$

The derivative of f and g are

$$f'(u) = \frac{1 \cdot (u+1)^2 - u \cdot 2(u+1)}{(u+1)^4} = \frac{u+1-2}{(u+1)^3} = \frac{u-1}{(u+1)^3},$$

and

$$g'(x) = \frac{1}{2\sqrt{x+1}}.$$

Hence the derivative of the composition is

$$h'(x) = \frac{d}{dx} \left\{ \frac{\sqrt{x+1}}{(\sqrt{x+1}+1)^2} \right\} = f'(u)g'(x) = \frac{u-1}{(u+1)^3} \cdot \frac{1}{2\sqrt{x+1}}.$$

The result should be a function of x , and we achieve this by replacing all u 's with $u = \sqrt{x+1}$

$$\frac{d}{dx} \left\{ \frac{\sqrt{x+1}}{(\sqrt{x+1}+1)^2} \right\} = \frac{\sqrt{x+1}-1}{(\sqrt{x+1}+1)^3} \cdot \frac{1}{2\sqrt{x+1}}.$$

The last step (where you replace u by its definition in terms of x) is important because the problem was presented to using x and y as variables while u was a variable you introduced.

Sometimes, it is possible to apply the Chain Rule without introducing a new letter. "the derivative is the derivative of the outside with respect to the inside times the derivative of the inside". Compute

$$\frac{d}{dx} \left(4 + \sqrt{7 + x^3} \right)$$

You could set $u = 7 + x^3$, and compute

$$\frac{d}{dx} \left(4 + \sqrt{7 + x^3} \right) = \frac{d(4 + \sqrt{u})}{du} \cdot \frac{du}{dx}$$

Instead of writing all this explicitly, you could think of $u = 7 + x^3$ as the function “inside the square root” and think of $(4 + \sqrt{u})$ as “the outside function”. You would then immediately write.

$$\frac{d}{dx} \left(4 + \sqrt{7 + x^3} \right) = \frac{1}{2\sqrt{7+x^3}} \cdot 3x^2.$$

3.6: The Chain Rule for computing more than two functions.

Often, we have to apply the Chain Rule more than once to compute a derivative. Thus if $y = f(u)$, $u = g(v)$, and $v = h(x)$ we have

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}.$$

In functional notation this is

$$\left. \frac{dy}{dx} \right|_{x=a} = \left. \frac{dy}{du} \right|_{u=c} \cdot \left. \frac{du}{dv} \right|_{v=b} \cdot \left. \frac{dv}{dx} \right|_{x=a}.$$

For example, if $y = \frac{1}{1 + \sqrt{9 + x^2}}$, then $y = \frac{1}{1 + u}$ where $u = 1 + \sqrt{v}$ and $v = 9 + x^2$ so

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = - \frac{1}{(1+u)^2} \cdot \frac{1}{2\sqrt{v}} \cdot 2x.$$

so

$$\left. \frac{dy}{dx} \right|_{x=4} = \left. \frac{dy}{du} \right|_{u=6} \cdot \left. \frac{du}{dv} \right|_{v=25} \cdot \left. \frac{dv}{dx} \right|_{x=4} = - \frac{1}{7} \cdot \frac{1}{10} \cdot 8.$$

Exercises:

i. Let $y = \sqrt{1 + x^3}$ and find dy/dx using the Chain Rule.

- ii. repeat the previous exercise with $y = (1 + \sqrt{1 + x})^3$.
- iii. Let $y = u^3 + 1$ and $u = 3x + 7$. Find $\frac{dy}{dx}$ and $\frac{dy}{du}$. Express the former in terms of x and the latter in terms of u .
- iv. Suppose that $f(x) = \sqrt{x}$, $g(x) = 1 + x^2$, $v(x) = f \circ g(x)$, $w(x) = g \circ f(x)$. Find formulas for $v(x)$, $w(x)$, $v'(x)$ and $w'(x)$.
- v. Compute the following derivatives
- (a) $f(x) = \sin \sin 2x - \cos \cos 3x$; (b) $f(x) = \sin \sin \frac{\pi}{x}$; ©
- $f(x) = \sin(\cos \cos 3x)$;
- (d) $f(x) = \frac{\sin \sin x^2}{x^2}$; (e) $f(x) = \tan \tan \sqrt{1 + x^2}$; (f)
- $f(x) = \cos^2 x - \cos \cos x^2$; (g) Find the derivative of
- $f(x) = x \cos \cos \frac{\pi}{x}$; (h) Suppose that
- $f(x) = x^2 + 1$, $g(x) = x + 5$ and $v = f \circ g$, $w = g \circ f$, $p = f \cdot g$, $q = g \cdot f$.
- Find $v(x)$, $w(x)$, $p(x)$ and $q(x)$.

3.7: Exponential growth and decay

A quantity X which depends on time t is said to grow or decay exponentially if it is given by

$$X(t) = X_0 e^{kt}. \quad (21)$$

The constant X_0 is the value of $X(t)$ at $t = 0$ (sometimes called “the initial value of X ”).

The derivative of an exponentially growing quantity, i.e. its rate of change with time, is given by $X'(t) = X_0 k e^{kt}$ so that

$$\frac{dX(t)}{dt} = kX(t) \quad (22)$$

In words, for an exponentially growing quantity the rate of change is always proportional to the quantity itself. The proportionality constant is k and is sometimes called “the relative growth rate”.

The property of exponential functions completely describes them, by which I mean that any function which satisfies (21) and (22). Suppose you have a function $X(t)$ for which $X'(t) = kX(t)$ holds at all times t . Then

$$\begin{aligned} \frac{d(X(t)e^{-kt})}{dt} &= k(t) \frac{de^{-kt}}{dt} + \frac{dX(t)}{dt} e^{-kt} \\ &= -kX(t)e^{-kt} + X'(t)e^{-kt} \\ &= (X'(t) - kX(t)) e^{-kt} \\ &= 0. \end{aligned}$$

It follows that $X(t)e^{-kt}$ does not depend on t . At $t = 0$ one has

$$X(t)e^{-kt} = X(0)e^0 = X_0$$

and therefore, we have

$$X(t)e^{-kt} = X_0 \text{ for all } t.$$

Multiply with e^{kt} and we end up with

$$X(t) = X_0 e^{kt}.$$

3.8: Half time and doubling time.

If $X(t) = X_0 e^{kt}$ then one has

$$X(t + T) = X_0 e^{kt+kT} = X_0 e^{kt} e^{kT} = e^{kT} X(t).$$

In words, after time T goes by an exponentially growing (decaying) quantity changes by a factor e^{kT} . If $k > 0$, so that the quantity is actually growing, then one calls

$$T = \frac{\ln \ln 2}{k}$$

In doubling time for X because $X(t)$ changes by a factor $e^{kT} = e^{\ln \ln 2} = 2$ every T times units: $X(t)$ doubles every T units.

If $k < 0$ then $X(t)$ is decaying and one calls

$$T = \frac{\ln \ln 2}{-k}$$

The half-life because $X(t)$ is denoted by a factor $e^{kT} = e^{\ln \ln 2} = \frac{1}{2}$ every T time units.

Determining X_0 and k . The general exponential growth/decay function (14) above contains only two constants, X_0 and k , and if you know the values of $X(t)$ at two different times then you can compute these constants.

Suppose that you know

$$X_1 = X(t_1) \text{ and } X_2 = X(t_2).$$

Then we have

$$X_0 e^{kt_1} = X_1 \text{ and } X_0 e^{kt_2} = X_2$$

in which t_1, t_2, X_1, X_2 are given and k and X_0 are known. One first finds k from

$$\frac{X_1}{X_2} = \frac{X_0 e^{kt_1}}{X_0 e^{kt_2}} = e^{k(t_1 - t_2)} \Rightarrow \ln \ln \frac{X_1}{X_2} = k(t_1 - t_2)$$

which implies

$$k = \frac{\ln \ln X_1 - \ln \ln X_2}{t_1 - t_2}$$

once you have computed k you can find X_0 from

$$X_0 = \frac{X_1}{e^{kt_1}} = \frac{X_2}{e^{kt_2}}.$$

(both expressions should give the same result.)

Conclusion

In this session, we covered various differentiation rules, including the Sum, Product, and Quotient rules, as well as the Power Rule. We also explored the Chain Rule, which allows us to find derivatives of compositions of functions. The Chain Rule is a powerful tool that enables us to differentiate more complex functions by breaking them down into simpler parts.

We saw how the Chain Rule works for differentiating functions involving trigonometric functions and exponential growth or decay. The Chain Rule allows us to find derivatives of functions involving multiple nested functions, making it an essential concept in calculus.

Overall, we learned how to apply these rules to find derivatives of various functions, which will be valuable in further exploring calculus and its applications in mathematics and science.

Summary

1. Differentiation rules include Sum Rule, Product Rule, and Quotient Rule.
2. Functions are denoted as $f(x)$ and $u(x)$ for differentiation purposes.
3. The Sum Rule states that the derivative of a sum is the sum of the derivatives of the individual terms.
4. The Product Rule states that the derivative of a product is the first function times the derivative of the second function plus the second function times the derivative of the first function.
5. The Quotient Rule states that the derivative of a quotient is the numerator's derivative times the denominator minus the number

Study session 4: Methods of Integration

Introduction

In this session, we will study the methods of integration, exploring various techniques to evaluate and solve integrals. Through this exploration, we will lay the groundwork for understanding advanced calculus concepts.

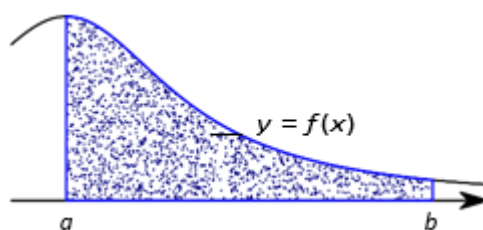
Learning Outcomes for Study Session 4:

When you have studied this session, you should be able to:

- 4.1. Define and use correctly all of the keywords printed in bold.
- 4.2. Explain the concept of integration and its significance in mathematics.
- 4.3. Understand and apply various notations used in integration.
- 4.4. Solve basic integration problems using different integration techniques. Identify real-world applications where integration is used to solve problems.

4.1: Methods of Integration

An integral of a function on some interval $[a, b]$. The most common interpretation of the integral is in terms of the area under the graph of the given function, so that is where we begin.



Let f be a function which is defined on some interval $a \leq x \leq b$ and assume it is positive, i.e. assume that its graph lies above the x axis. How large is the area of the region caught between the x axis, the graph of $y = f(x)$ and the vertical lines $y = a$ and $y = b$.

You can try to compute this area by approximating the region with many thin rectangles. To make the approximating region you choose a particular of the interval $[a, b]$, i.e you pick numbers $x_1 < \dots < x_n$ with

$$a = x_0 < x_1 < x_2, \dots, \Delta x_n = x_n - x_{n-1}$$

In each interval we choose point c_k i.e in the first interval we choose some number $x_{n-1} \leq c_1 \leq x_1$, in the second interval we choose $x_1 \leq c_2 \leq x_2, \dots$, and in the last interval we choose some number $x_{n-1} \leq c_n \leq x_n$.

We then define n rectangles: the base of the k^{th} rectangle is the interval $[x_{k-1}, x_k]$ on the x -axis, while its height is $f(x)$ (here k can be any integer from 1 to n .)

The area of the k^{th} rectangle is of course the product of the height and width, i.e., its area is $f(c_k)\Delta x_k$. Adding these we see that the total area of the rectangle is

$$R = f(c_1)\Delta x_1 + f(c_2)\Delta x_2 + \dots + f(c_n)\Delta x_n. \quad (23)$$

This kind of sum is called **Riemann sum**.

If the partition is sufficiently fine then one would expect this sum, i.e., the total area of all rectangles to be a good approximation of the area of the region under the graph. Replacing the partition by a finer partition, with more division points, should improve the approximation. So you would expect the area to be the limit of Riemann-sums like R “as the partition becomes finer and finer”. A precise formulation of the definition goes like this:

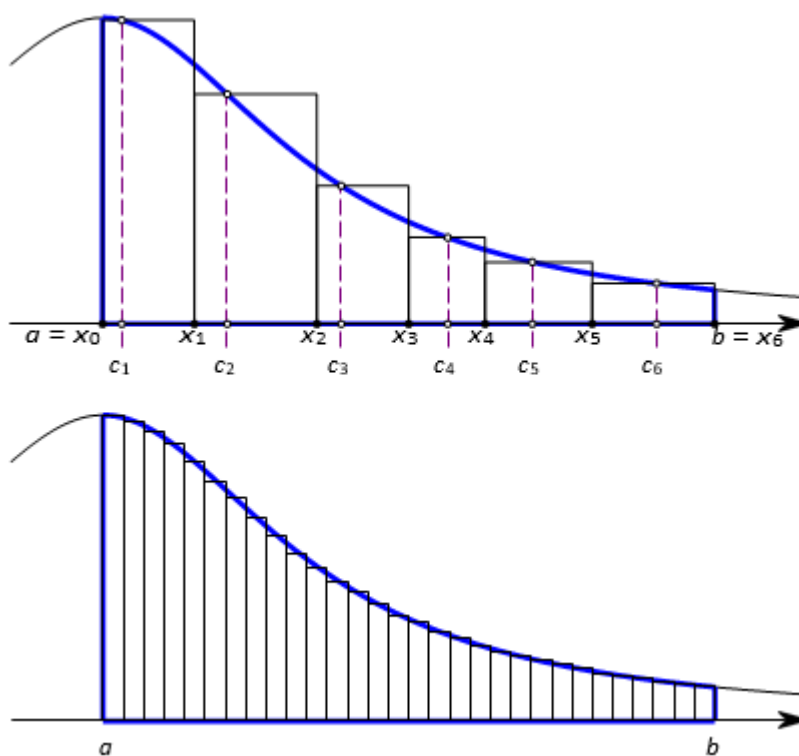


Figure 4.1: Riemann Sum for Approximating Area under curve

A Riemann sum in which the interval $a < x < b$ has been cut up into six smaller intervals. In each of those intervals a point c_1 has been chosen at random, and the resulting rectangles with heights $f(c_1) \dots f(c_6)$ were drawn. The total area under the graph of the function is roughly equal to the total area of the rectangles. BOTTOM. Refining the partition. After adding more partition points the combined area of the rectangles will be a better approximation of the area under the graph of the function f .

Definition. If f is a function defined on an interval $[a, b]$ then we say that

$$\int_a^b f(x)dx = I,$$

i.e., the integral of

$f(x)$ from $x = a$ to b equal I , if for $\varepsilon > 0$ one can find a $\delta > 0$ such that

$$|f(c_1)\Delta x_1 + f(c_2)\Delta x_2 + \dots + f(c_n)\Delta x_n - I| < \varepsilon$$

holds for every partition all of whose intervals have length $\Delta x_k < \delta$.

When f changes its sign

If the function f is not necessarily positive in the interval $a \leq x \leq b$, then we still define the integral in exactly the same way: as a way of Riemann sums whose mesh size becomes smaller and smaller. However, the interpretation of the integral as “the area of the region between the graph and the axis” has a twist to it.

Let f be some function on an interval $a \leq x \leq b$, and form the Riemann sum

$$R = f(c_1)\Delta x_1 + f(c_2)\Delta x_2 + \dots + f(c_n)\Delta x_n.$$

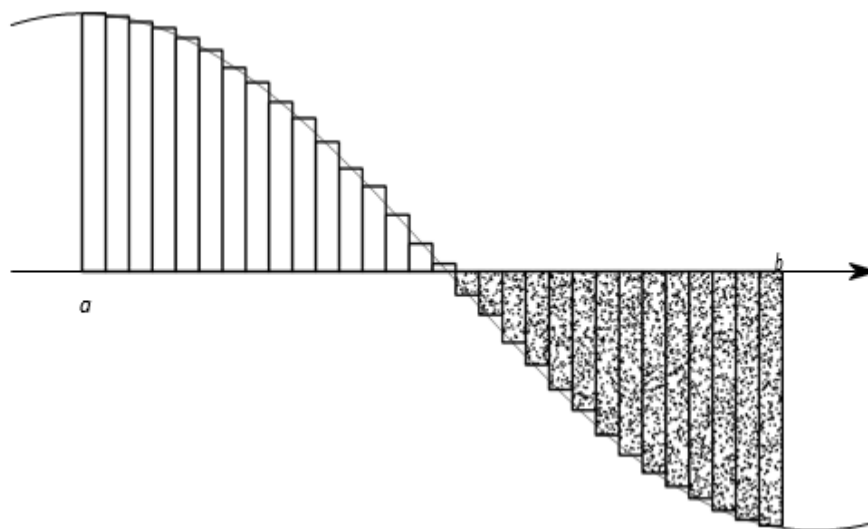


Figure 4.2: Riemann Sum for a Function with Sign Changes

Always remember this: **AREAS ARE ALWAYS POSITIVE NUMBERS.**

The Riemann-sum corresponding to this picture is the total area of the rectangles above the x -axis *minus* the total area of the rectangles below the x -axis.

When f can be positive or negative, then the terms in the Riemann sum can also be positive or negative. If $f(c_k)$ is a quantity $f(c_k)\Delta x_k$ is the area of the corresponding rectangle, but if $f(c_k) < 0$ then $f(c_k)\Delta x_k$ is a negative number, namely *minus* the area of the corresponding rectangle. The Riemann sum is therefore the area of the rectangles above the x -axis *minus* the area below the axis and above the graph.

Taking the limit over the finer and finer partitions, we conclude that

$$\int_a^b f(x)dx = \frac{\text{area above the } x\text{-axis, below the graph}}{\text{minus the area below the } x\text{-axis, above the graph}}$$

4.2 : The Fundamental Theorem of Calculus

Definition. A function is called an antiderivative of f on the interval $[a, b]$ if one has $F'(x) = f(x)$ for all x with $a < x < b$.

For instance, $F(x) = \frac{1}{2}x^2$ is an antiderivative of $f(x) = x$, but so is $G(x) = \frac{1}{2}x^2 + 2008$.

Theorem.

If f is a function whose $\int_a^b f(x)dx$ exists, and if F is an antiderivative of f on the interval $[a, b]$, then one has

$$\int_a^b f(x)dx = F(b) - F(a). \quad (24)$$

Because of this theorem the expression on the right appears so often that various abbreviations have been invented.

We will abbreviate

$$F(b) - F(a) = [F(x)]_{x=a}^b = [F(x)]_a^b.$$

Terminology.

In the integral

$$\int_a^b f(x)dx$$

the numbers a and b are called the bounds of the integral, the function $f(x)$ which is being integrated is called the integrand, and the variable x is the integration *variable*.

The integration variable is a *dummy variable*. If you systematically replace it with another variable, the resulting integral will still be the same. For instance,

$$\int_0^1 x^2 dx = \left[\frac{1}{3} x^3 \right]_{x=0}^1 = \frac{1}{3}.$$

and if you replace x by φ you still get

$$\int_0^1 \varphi^2 d\varphi = \left[\frac{1}{3} \varphi^3 \right]_{\varphi=0}^1 = \frac{1}{3}.$$

Another way to appreciate that the integration variable is a dummy variable is to look at the Fundamental Theorem again:

$$\int_a^b f(x) dx = F(b) - F(a).$$

The right-hand side tells you that the value of the integral depends on a and b , and has absolutely nothing to do with the variable x .

Exercise:

1. What is a Riemann sum of a function $y = f(x)$?
2. Let f be the function $f(x) = 1 - x^2$. Draw the graph of $f(x)$ with $0 \leq x \leq 2$.

Compute the Riemann-sum for the partition.

$0 < \frac{1}{3} < 1 < \frac{3}{2} < 2$ of the integral $[a, b] = [0, 2]$ if you choose each c_k to be left endpoint of the interval it belongs to. Draw the corresponding rectangles (add them to your drawing of the graph of f). Then compute

the Riemann-sum you get if you choose the c_k to be the right endpoint of the interval it belongs to. Make a new drawing of the graph of f and include the rectangles corresponding to the right endpoint Riemann-sum.

4.3: The Indefinite Integral

The fundamental theorem tells us that in order to compute the integral of some function f over an interval $[a, b]$ you should first find an antiderivative F of f . In practice, much of the effort required to find an integral goes into finding the antiderivative. In order to simplify the computation of the integral.

$$\int_a^b f(x)dx = F(b) - F(a) \quad (25)$$

the following notation is commonly used for the antiderivative:

$$F(x) = \int f(x)dx. \quad (26)$$

For instance,

$$\int x^2 dx = \frac{1}{3}x^3, \quad \int \sin 5x dx = -\frac{1}{5}\cos 5x, \text{ etc...}$$

The integral appear here does not have the integration bounds a and b . It is called an ***indefinite integral***, as opposed to the integral in (16) which is called a ***definite integral***. You use the indefinite integral if you expect the computation of the antiderivative to be a lengthy affair, and you do not want to write the integration bounds a and b all the time.

To properly understand the concept of integrals, it is crucial to distinguish between the two types. Table 4.1 below provides a comprehensive comparison, outlining the key differences between the two kinds of integrals.

Table 4.1: Definite and Indefinite Integrals

Indefinite Integral	Definite Integral
$\int f(x)$ is a function of x	$\int_a^b f(x)dx$ is a number.
By definition $\int f(x)dx$ is any function of x whose derivative is $f(x)$	$\int_a^b f(x)dx$ was defined in terms of Riemann sums and can be interpreted as “area under the graph of $y = f(x)$ ” at least when $f(x) > 0$.
x is not a dummy variable for example $\int 2x dx = x^2 + C$ and $\int 2t dt = t^2 + C$	x is a dummy variable, for example $\int_0^1 2x dx = 1$, and $\int_0^1 2t dt = 1$, so $\int_1^1 2x dx$
are functions of different variables, so they are not equal	

You can always check the answer. Suppose you want to find out an approximation of a given function $f(x)$. You can then throw away the dubious computation which you don’t really trust you get an “answer”, $F(x)$ turns out to

be equal to $f(x)$, then your $F(x)$ is indeed an antiderivative and your computation isn't important anymore.

For example, suppose that we want to find $\int \ln \ln x dx$. My cousin Louie says it might be $F(x) = x \ln \ln x$.

$$\frac{d}{dx}(x \ln \ln x - x) = x \cdot \frac{1}{x} + 1 \cdot \ln \ln x - 1 = \ln \ln x.$$

But $\int \ln \ln x dx = x \ln \ln x - x + C$.

Let a function defined on some interval $a \leq x \leq b$. If $F(x)$ is an antiderivative of $f(x)$ on the interval, then for any C the function $\tilde{F}(x) = F(x) + C$ will also be an antiderivative of $f(x)$. So one given function $f(x)$ has many different antiderivatives, obtained by adding different constants to one given antiderivatives.

Theorem. If $F_1(x)$ and $F_2(x)$ are antiderivatives of the same function $f(x)$ on some interval $a \leq x \leq b$, then there is a constant C such that $F_1(x) = F_2(x) + C$.

Proof.

Consider the difference

$G(x) = F_1(x) - F_2(x)$. Then $G'(x) = F_1'(x) - F_2'(x) = f(x) - f(x) = 0$, so that $G(x)$ must be constant. Hence $F_1(x) - F_2(x) = C$ for some constant.

It follows that there is ambiguity in the notation $\int f(x)dx$. Two functions $F_1(x)$ and $F_2(x)$ can both equal $\int f(x)dx$ without equaling each other. When this happens, they (F_1 and F_2) differ by a constant. This can sometimes lead to confusing situations, e.g. you can check that

$$\int 2 \sin x \cos x dx = \sin^2 x$$

$$\int 2 \sin x \cos x dx = -\cos^2 x$$

are both correct. (Just differentiate the two functions $\sin^2 x$ and $-\cos^2 x$). These two answers look different until you realise that because of the trig identity $\sin^2 x + \cos^2 x = 1$ they really differ by a constant: $\sin^2 x$ and $-\cos^2 x + 1$.

4.4: Properties of the integral

Just as we had a list of properties for the limits and derivatives of sums and products of functions, the integral has similar properties.

Suppose we have two functions $f(x)$ and $g(x)$ with antiderivatives $F(x)$ and $G(x)$, respectively. Then we know that

$$\frac{d}{dx}\{F(x) + G(x)\} = F'(x) + G'(x) = f(x) + g(x),$$

in words, $F + G$ is an antiderivatives of $f + g$, which we can write as

$$\int \{F(x) + G(x)\}dx = \int f(x)dx + \int g(x)dx \quad (27)$$

Similarly, $\frac{d}{dx}(cF(x)) = cF'(x) = cf(x)$ implies that

$$\int cf(x)dx = c \int f(x)dx \quad (28)$$

If c is a constant

These properties imply analogous properties for the definite integral. For any pair of function on an interval $[a, b]$ one has

$$\int_a^b [f(x) + g(x)]dx = \int_a^b f(x)dx + \int_a^b g(x)dx, \quad (29)$$

and for any function f and constant c one has

$$\int_a^b cf(x)dx = c \int_a^b f(x)dx. \quad (30)$$

Definite integrals have one other property for which there is no analog in indefinite integrals: if you split the interval of integration into two parts, then the integral over the whole is the sum of the integrals over the parts. The following theorem says it more precisely.

Theorem. Given $a < c < b$, and a function on the interval $[a, b]$ then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx \quad (32)$$

Proof: Let F be an antiderivative of f . Then

$$\int_a^c f(x)dx = F(c) - F(a) \text{ and } \int_c^b f(x)dx = F(b) - F(c),$$

So that

$$\begin{aligned}\int_a^b f(x)dx &= F(b) - F(a) \\ &= F(b) - F(c) + F(c) - F(a) \\ &= \int_a^c f(x)dx + \int_c^b f(x)dx\end{aligned}$$

So far we have always assumed that $a < b$ in all indefinite integrals $\int_a^b \dots$. The fundamental theorem suggests that when $b < a$, we should define the integral as

$$\int_a^b f(x)dx = F(b) - F(a) = - (F(a) - F(b)) = - \int_b^a f(x)dx \quad (32)$$

For instance,

$$\int_1^0 xdx = - \int_0^1 xdx = - \frac{1}{2}$$

4.5: The definite integral as a function of its integration bounds

Consider the expression

$$I = \int_0^x t^2 dt$$

What does I depend on? To see this, you calculate the integral and you find

$$I = \left[\frac{1}{3}t^3 \right]_0^x = \frac{1}{3}x^3 - \frac{1}{3}0^3 = \frac{1}{3}x^3$$

So, the integral depends on x . It does not depend on t , since t is a “dummy variable”. (In this way you can use integrals to define new functions. For instance, we could define

$$I(x) = \int_0^x t^2 dt$$

Which would be a roundabout way of defining the function $I(x) = \frac{x^3}{3}$. Again, since t is a dummy variable, we can replace it by any other variables we like. Thus

$$I(x) = \int_0^x \alpha^2 d\alpha$$

Defines the same function (*namely*, $I(x) = \frac{1}{3}x^3$).

The previous example does not define a new function $I(x) = \frac{x^3}{3}$. An example of a new function defined by an integral is the “error-function” from statistics. It is given by

$$\operatorname{erf}(x) \stackrel{\text{def}}{=} \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (33)$$

So $\operatorname{erf}(x)$ is the area of the shaded region in figure 4.3.

The integral in (33) cannot be computed in terms of the standard functions (square and higher root, sine, cosine, exponential and logarithms). Since the integral in (33) occurs very often in statistics (in relation with the so-called normal distribution) it has been given a name, namely, “ $\operatorname{erf}(x)$ ”.

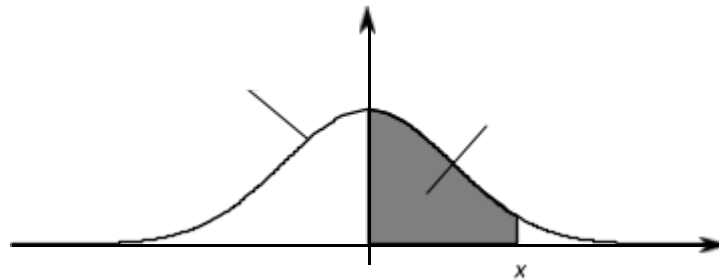


Figure 4.3: Shaded Region Representing erfx

How do you differentiate a function that is defined by an integral? The answer is simply, for if $f(x) = F'(x)$ then the fundamental theorem says that

$$\int_a^x f(t) dt = F(x) - F(a)$$

and therefore

$$\frac{d}{dx} \int_a^x f(t) dt = \frac{d}{dx} \{F(x) - F(a)\} = F'(x) = f(x),$$

i.e.,

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Similar calculation gives you

$$\frac{d}{dx} \int_x^b f(t) dt = -f(x)$$

So, what is the derivative of the error function? We have

$$\begin{aligned} \text{erf}'(x) &= \frac{d}{dx} \left\{ \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \right\} \\ &= \frac{2}{\sqrt{\pi}} \frac{d}{dx} \int_0^x e^{-t^2} dt \end{aligned}$$

$$= \frac{2}{\sqrt{\pi}} e^{-x^2}$$

4.6: Method of Substitution

The chain rule states that

$$\frac{dF(G(x))}{dx} = F'(G(x)) \cdot G'(x)$$

Such that

$$\int F'(G(x)) \cdot G'(x) dx = F(G(x)) + C$$

4.6.1: Example. Consider the function $f(x) = 2x \sin(x^2 + 3)$. It does not appear in the list of standard antiderivatives we know by heart. But we do notice that $2x = \frac{d}{dx}(x^2 + 3)$. So let's call $G(x) = x^2 + 3$, and $f(u) = -\cos u$, then

$$F(G(x)) = -\cos(x^2 + 3)$$

and

$$\frac{dF(G(x))}{dx} = \sin(x^2 + 3) \cdot 2x = f(x)$$

So that

$$\int 2x \sin(x^2 + 3) dx = -\cos(x^2 + 3) + C \quad (34)$$

4.6.2: Leibniz' notation for substitution.

The most transparent way of computing an integral by substitution is by following Leibniz and introducing new variables. Thus, the integral

$$\int f(G(x)) G'(x) dx$$

where $f(u) = F'(u)$, introduces the substitution $u = G(x)$, and agree to write

$$du = dG(x) = G'(x)dx$$

Then we get

$$\int f(G(x))G'(x)dx = \int f(u)du = F(u) + C$$

At the end of the integration, we must remember that u really stands for $G(x)$, so that

$$\int f(G(x))G'(x)dx = F(x) + C = F(G(x)) + C$$

As an example, let's do the integral 58 using Leibniz' notation. We want to find

$$\int 2x \sin(x^2 + 3)dx$$

and decide to substitute $z = x^2 + 3$ (the substitution variable does not always have to be called u). Then we compute

$$dz = d(x^2 + 3) = 2xdx \text{ and } \sin(x^2 + 3) = \sin z$$

So that

$$\int 2x \sin(x^2 + 3)dx = \int \sin z dz = -\cos z + C$$

Finally, we get rid of substitution variable z , and we find

$$\int 2x \sin(x^2 + 3) = -\cos(x^2 + 3) + C$$

When we do integrals in this calculus class, we always get rid of the substitution variable because it is a variable we invented, and which does not

appear in the original problem. But if you are doing an integral which appears in some longer discussion of real-life (or real-lab) situation, then it may be that the substitution variable actually has a meaning (e.g., “the effective stoichiometric modality of CQF self-inhibition”) in which case you may want to strip the last set and leave the integral in terms of the (meaningful) substitution variable.

4.5.3. Substitution for Definite Integrals.

For definite integrals the chain rule

$$\frac{d}{dx}(F(G(x))) = F'(G(x))G'(x) = f(G(x))G'(x)$$

Implies

$$\int_a^b f(G(x))G'(x)dx = F(G(b)) - F(G(a))$$

Which you can also write as

$$\int_{x=a}^b f(G(x))G'(x)dx = \int_{u=G(a)}^{G(b)} f(u)du \quad (35)$$

4.5.4. Example of Substitution in a Definite Integral.

let's compute

$$\int_0^1 \frac{x}{1+x^2} dx$$

Using the substitution $u = G(x) = 1 + x^2$. Since $du = 2xdx$, the associated **indefinite** integral is

$$\int \frac{1}{1+x^2} x dx \stackrel{u=\frac{1}{x}}{\underset{\frac{1}{u}}{=}} \frac{1}{2} \int \frac{1}{u} du$$

To find the definite integral you must compute the new integration bounds $G(0)$ and $G(1)$ (see equation (59).) If x runs between $x = 0$ and $x = 1$, then $G(x) = 1 + x^2$ runs between $u = 1 + 0^2 = 1$ and $u = 1 + 1^2 = 2$, so the indefinite integral we must compute is

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \int_1^2 \frac{1}{u} du \quad (36)$$

which is our list of memorable integrals. So, we find

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \int_1^2 \frac{1}{u} du = \frac{1}{2} [\ln u]_1^2 = \frac{1}{2} \ln 2$$

Sometimes the integrals in (36) are written as

$$\int_{x=0}^1 \frac{x}{1+x^2} dx = \frac{1}{2} \int_{u=1}^2 \frac{1}{u} du$$

to emphasise (and remind yourself) to which variables the bounds in the integral refer.

Exercises:

Compute these derivatives:

1. $\frac{d}{dx} \int_0^x (1 + t^2)^4 dt$

2. $\frac{d}{dx} \int_x^1 \ln z dz$

3. $\frac{d}{dt} \int_0^t \frac{dx}{1+x^2}$

4. $\frac{d}{dt} \int_0^{\frac{1}{t}} \frac{dx}{1+x^2}$

5. $\frac{d}{dt} \int_x^{2x} s^2 ds$

6. $\frac{d}{dx} \int_0^t \frac{dx}{1+x^2}$

7. $\frac{d}{dq} \int_{-q}^q \frac{dx}{1+x^2}$ (which values of q are allowed here?)

8. $\frac{d}{dx} \int_0^{t^2} e^{2x} dx$

9. **Group Problem**

Compute the following indefinite integrals:

10. $\int \{6x^5 - 2x^{-4} - 7x\} dx$

11. $\int_1^4 x^{-2} dt$

4.7: Applications of the Integral

The integral appears to be the answer to many questions. In this section, we will describe a number of “things which are integral.” In each example there is a quantity we want to compute, and which we can approximate through Riemann-sums. After letting the partition become arbitrarily fine, we then find that the quantity we are looking for is given by an integral. The derivations are an important part of the subject.

4.7.1. Area between graphs

Suppose you have two functions f and g on an interval $[a, b]$, one of which is always larger than the other, i.e., for which you know that $f(x) \leq g(x)$ for all x in the interval $[a, b]$. Then the area of the region between the graphs of the two functions

$$Area = \int_a^b (g(x) - f(x)) dx \quad (37)$$

To get this formula you approximate the region by a large number of thin rectangles. Choose a partition $a = x_0 < x_1 < \dots < x_n = b$ of the interval $[a, b]$; choose a number c_k in each interval $[x_{k-1}, x_k]$; form the rectangles

$$x_{k-1} \leq x_k, \quad f(c_k) \leq g(c_k)$$

The area of this rectangle is

$$width \times height = \Delta x_k \times (g(c_k) - f(c_k))$$

hence the combined area of the rectangles is

$$R = (g(c_1) - f(c_1))\Delta x_1 + \dots + (g(c_n) - f(c_n))\Delta x_n$$

which is just the Riemann-sum for the integral

$$I = \int_a^b (g(x) - f(x))dx$$

So,

1. Since the area of the region between the graphs of f and g is the limit of the combined area of the rectangles

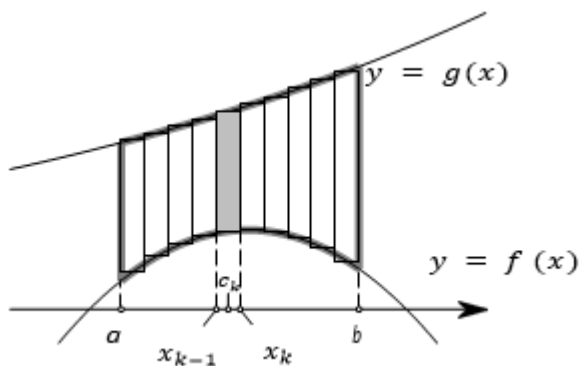


Figure 4.4. Finding the area between two graphs using Riemann-sums.

2. and since this combined area is equal to the Riemann sum R ,
 3. and since the Riemann-sum R converge to the integral I
- we conclude that the area between the graphs of f and g is exactly the integral I

Exercises:

1. find the area of the region bounded by the parabola $y^2 = 4x$ and the line $y = 2x$
2. Find the area bounded by the curve $y = x(x - 2)$ and the line $x = 2y$
3. find the area bounded by the curve $x^2 = 4y$ and the line $x = 4y - 2$
4. Calculate the area of the region included between the parabolas $y = x^2$ and $x = y^2$
5. Find the area of the region included between the parabola $y^2 = x$ and the line $x + y = 2$.
6. Find the area of the region bounded by the curves $y = \sqrt{x}$ and $y = x$
7. Use integration to find the area of the triangular region bounded by the lines $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.
8. Find the area bounded by the parabola $x^2 - 2 = y$ and the line $x + y = 0$
9. Where do the graphs of $f(x) = x^2$ and $g(x) = \frac{3}{(2+x^2)}$ intersect? Find the area of the region which lies above the graph of g and below the graph of f . (Hint: if you need to integrate $\frac{1}{(2+x^2)}$ you could substitute $x = u\sqrt{2}$)
10. graph the curve $y = \left(\frac{1}{2}\right)x^2 + 1$ and the straight line $y^2 = x$ and $x^2 = 16y$

4.7.2: Cavalieri's principle and volumes of solid

You can use integration to derive the formulas for volumes of spheres, cylinder, cones, and many more solid objects in a systematic way. In this section we'll see the "method of slicing"

Example-Volume of Pyramid. As an example, let's compute the volume of a pyramid whose base is square of side 1, and whose height is 1. Our strategy will be to divide the pyramid into thin horizontal slices whose volumes we can compute, and to add the volume of the slices to get the volume of the pyramid. To construct the slices, we choose a partition of the (height) interval $[0, 1]$ into N subintervals, i.e., we pick numbers

$$0 = x_0 < x_1 < x_2 < \dots < x_{N-1} < x_N = 1$$

and as usual we set $\Delta x_k = x_k - x_{k-1}$, we define the mesh size of the partition to be the largest of the Δx_k .

The k^{th} slice consists of those points on the pyramid whose height is between x_{k-1} and x_k . The intersection of the pyramid with the plane at height x is a square, and by similarity the length of the side of this square is $1 - x$. Therefore the bottom of the k^{th} slice is square with side $1 - x_{k-1}$, and its top is a square with side $1 - x_k$. The height of the slice is $x_k - x_{k-1} = \Delta x_k$.

Thus the k^{th} slice contains a block of height of the slice is Δx_k whose base is a square side $1 - x_{k-1}$, and its volume must therefore be larger than

$(1 - x_k)^2 \Delta x_k$. On the other hand the k^{th} slice is contained in a bloc of the same height whose base is a square with side $1 - x_{k-1}$. The volume of the slice is therefore not more than $(1 - x_k)^2 \Delta x_k$. So we have

$$(1 - x_k)^2 \Delta x_k < \text{volume of } k^{th} \text{ slice} < (1 - x_k)^2 \Delta x_k$$

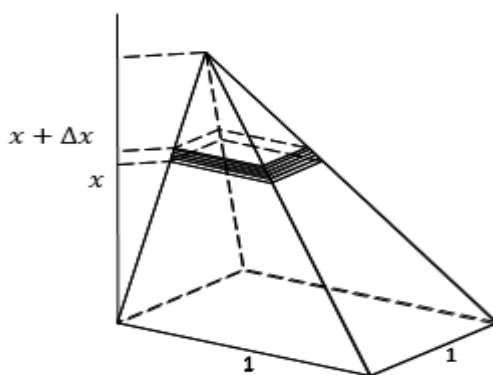


Figure 4.5. The slice at height x is a square with side $1 - x$.

Therefore, there is some c_k in the interval $[x_{k-1}, x_k]$ such that

$$\text{volume of } k^{th} \text{ slice} = (1 - c_k)^2 \Delta x_k$$

adding the volume of the slices we find that the volume V of the pyramid is given by

$$(1 - c_1)^2 \Delta x_1 + \dots (1 - c_N)^2 \Delta x_N$$

The right-hand side in this equation is a Riemann sum for the integral

$$I = \int_0^1 (1 - x)^2 dx$$

and therefore, we have

$$I = \left\{ (1 - c_1)^2 \Delta x_1 + \dots (1 - c_N)^2 \Delta x_N \right\} = V$$

Compute the integral and you find that the volume of the pyramid is

$$V = \frac{1}{3}$$

4.7.3: General case.

The “method of slicing” which we just used to compute the volume of a pyramid works for solids of any shape. The strategy always consists of dividing the solid into many thin (horizontal) slices, compute their volumes, and recognize that the total volume of the slices is a Riemann sum for some integral. That integral then is the volume of the solid.

To be more precise, let a and b be the lowest and highest points on the solid, and let $a = x_0 < x_1 < x_2 < \dots < x_{N-1} < x_N = b$ be a partition of the interval $[a, b]$. Such a partition divides the solid into N distinct slices, where slice number k consist of all points in the solid whose highest is between x_{k-1} and x_k . The thickness of k^{th} slice $\Delta x_k = x_k - x_{k-1}$. If

$A(x)$ = area of the intersection of the solid with the plane at height x

Then we can approximate the volume of k^{th} slice by

$$A(c_k) \Delta x_k$$

Where c_k is any number (height) between x_{k-1} and x_k

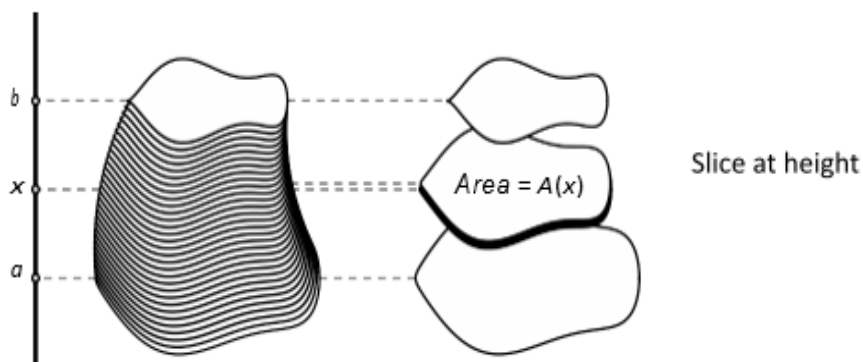


Figure 4.6. Slicing a solid to compute its volume.

The volume of one slice is approximately the product of its thickness (Δx) and the area $A(x)$ of its top. Summing the volume $A(x)\Delta x$ over all slices leads

The total volume of all slices is therefore approximately

$$V \approx A(c_1)\Delta x_1 + \dots A(c_N)\Delta x_N$$

While this formula only holds approximately, we expect the approximation to get better as we make the partition finer, and thus

$$V = \{A(c_1)\Delta x_1 + \dots A(c_N)\Delta x_N\} \quad (38)$$

On the other hand, the sum on the right is a Riemann sum for the integral

$I = \int_a^b A(x)dx$, so the limit is exactly this integral. therefore, we have

$$V = \int_a^b A(x)dx \quad (39)$$

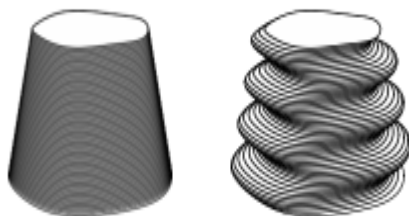


Figure 4.6. Cavalieri's principle.

Both solids consist of a pile of horizontal slices. The solid on the right was obtained from the solid on the left by sliding some of the slices to the left and others to the right. This operation does not affect the volumes of the slices, and hence both solids have the same volume.

4.7.4: Cavalieri's Principle.

The formula (38) for the volume of a solid which we have just derived shows that the volume only depends on the area $A(x)$ of the cross sections of the solid, and not on the particular shape these cross sections may have. This observation is older than calculus itself and goes back at least to Bonaventura Cavalieri (1598-1647) who said: if the intersections of two solids with a horizontal plane always have the same area, no matter what the height of the horizontal plane may be, then the two solids have the same volume

This principle is often illustrated by considering a stock of coins: if you put a number of coins on top of each other then, the total volume of the coins is just the sum of the volumes of the coins. If you change the shape of the pile by sliding the coins horizontally then the volume of the pile will still be the sum of the volumes of the coins, e.i. it doesn't change.

4.7.5: Solid of revolution.

In principle, formula (38) allows you to compute the volume of any solid, provided you can compute the area $A(x)$ of all cross sections. One class of solids for which the areas of the cross sections are easy are the so-called “solids of revolution.”

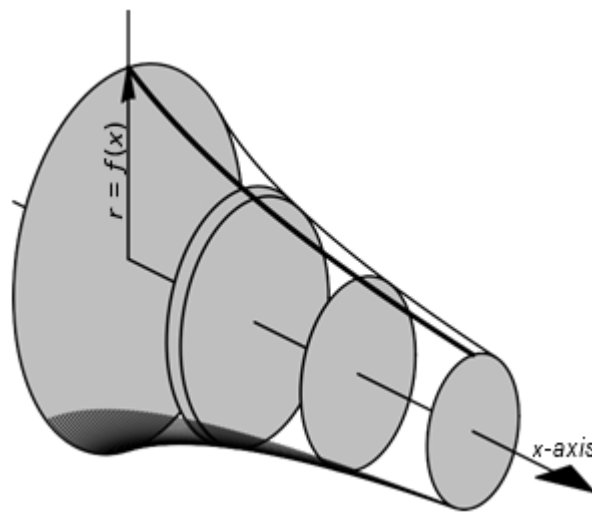


Figure 4.7: Solid of Revolution - Rotation of a Positive Function

A solid of revolution consists of all points in three-dimensional space whose distance r to the x -axis satisfies $r \leq f(x)$.

A solid of revolution is created by rotating (revolving) the graph of a positive function around the x – axis. More precisely, let f be a function which is defined on an interval $[a, b]$ and which is always positive ($f(x) > 0$ for all x). If you now imagine the x – axis floating in three-dimensional space, then the solid of revolution obtained by rotating the graph of f around the x – axis consists of all points in three-dimensional space with $a \leq x \leq b$, and whose distance to the x – axis is no more than $f(x)$.

Yet another way of describing the solid of revolution is to say that the solid is the union of all discs which meet the $x = \text{axis}$ perpendicularly and whose radius is given by $r = f(x)$.

If we slice the solid with planes perpendicular to the $x - \text{axis}$, then (63) tells us the volume of the solid.

Each slice is a disc of radius $r = f(x)$ so that its area is $A(x) = \pi r^2 = \pi f(x)^2$. We therefore find that

$$V = \pi \int_a^b f(x)^2 dx \quad (40)$$

4.8: Examples of Volumes of solids of revolution

problem 1: Revolve R around the $y - \text{axis}$. Consider the solid obtained by revolving the region

$$R = \{x, y \mid 0 \leq x \leq 2, (x - 1)^2 \leq y \leq 1\}$$

Around the $y - \text{axis}$

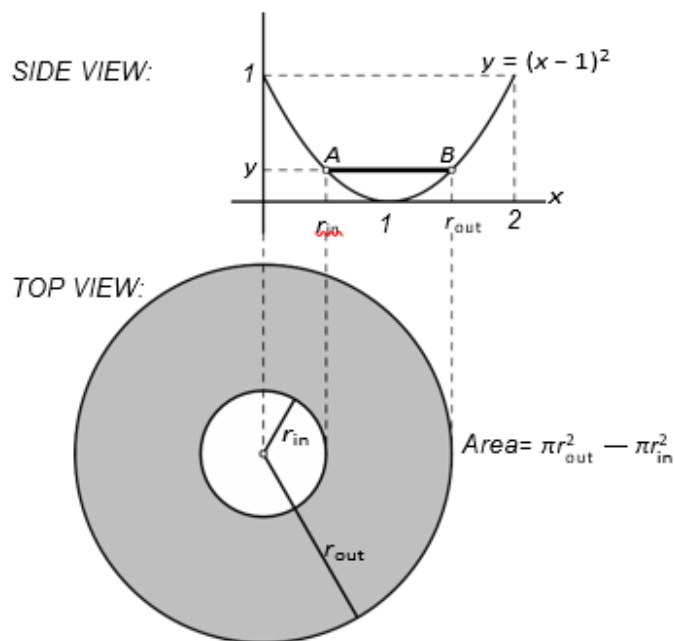
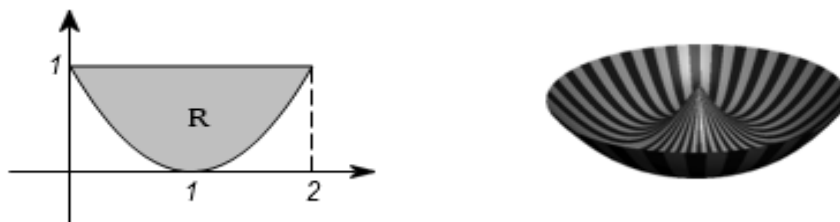


Figure 4.8: Volume of Revolved Solid with Washer Cross Sections

Computing the volume of the solid you get when you revolve the region R around the $y - axis$.

A horizontal cross section of the solid is a “washer” with inner radius r_{in} and outer radius r_{out}



Solution: the region we have to revolve around the $y - axis$ consists of all points above the parabola $y = (x - 1)^2$ but below the line $y = 1$

If we intersect the solid with a plane at height y then we get a ring-shaped region, or “annulus”, i.e. a large disc with a smaller disc removed. You can see it in the figure below: if you cut the region R horizontally at height y you get the line segment AB , and if you rotate this segment around the y – axis you get the grey ring region pictured below the graph. Call the radius of the outer circle r_{out} and the radius of the inner circle r_{in} . These radii are the two solutions of

$$y = (x - 1)^2$$

so they are:

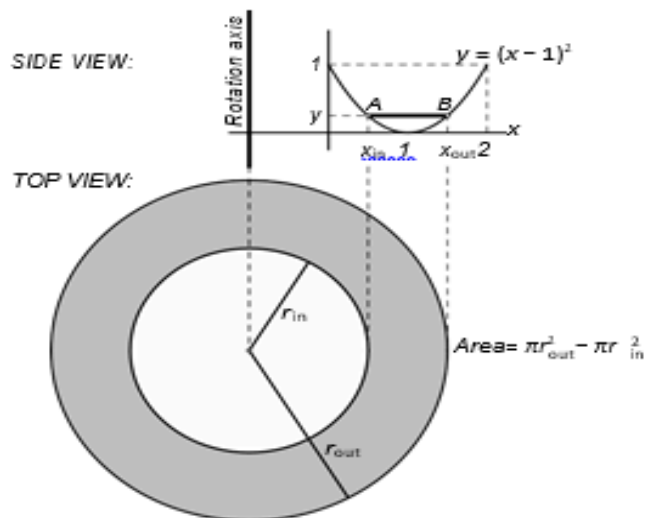
$$r_{in} = 1 - \sqrt{y}, \quad r_{out} = 1 + \sqrt{y}$$

The area of the cross section is therefore given by

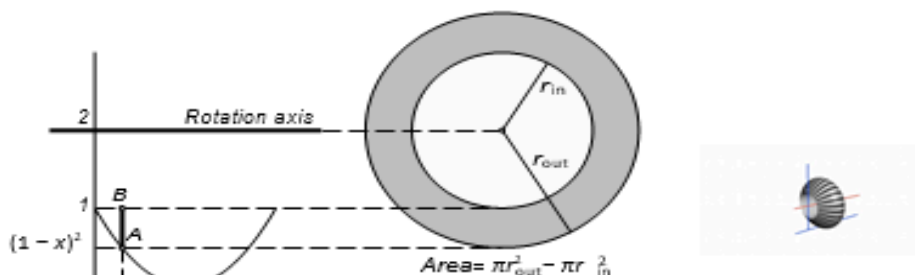
$$A(y) = \pi r_{out}^2 - \pi r_{in}^2 = \pi(1 + \sqrt{y})^2 - \pi(1 - \sqrt{y})^2 = 4\pi\sqrt{y}$$

The y – values which occur in the solid are $0 \leq y \leq 1$ and hence the volume of the solid is given by

$$V = \int_0^1 A(y) dy = 4\pi \int_0^1 \sqrt{y} dy = 4\pi \times \frac{2}{3} = \frac{8\pi}{3}$$



Problem 2: Rotate the line segment AB around the vertical line $x = -1$ and you get a washer.



Problem 3: If you rotate the line segment AB around the horizontal line $y = 2$ you once again get a washer.

Figure 4.9: Volume of revolution

Problem 2: Revolve R around the line $x = -1$.

Find the volume of the solid of revolution obtained by revolving the same region R around the line $x = -1$.

Solution: The line $x = -1$ is vertical, so we slice the solid with horizontal planes. The height of each plane will be call y .

As before the slices are ring shaped region but the inner and out radii are now given by

$$r_{in} = 1 + x_{in} = 2 - \sqrt{y}, \quad r_{out} = 1 + x_{out} = 2 + \sqrt{y}$$

The volume is therefore given by

$$V = \int_0^1 (\pi r_{out}^2 - \pi r_{in}^2) dy = \pi \int_0^1 8\sqrt{y} dy = \frac{16\pi}{3}$$

Problem 3: Revolve R around the line $y = 2$.

Compute the volume of the solid you get when you revolve the same region R around the line $y = 2$.

Solution:

This time the line around which we rotate R is horizontal, so we slice the solid with planes perpendicular to the x - axis.

A typical slice is obtained by revolving the segment AB about the line $y = 2$.

The result is again an annulus, and from the figure we see that the inner and outer radii of the annulus are

$$r_{in} = 1, \quad r_{out} = 2 - (1 - x)^2$$

The area of the slice is therefore

$$A(x) = \pi \{2 - (1 - x)^2\}^2 - \pi 1^2 = \pi \{3 - 4(1 - x)^2 + (1 - x)^4\}$$

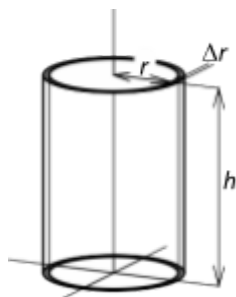
The x values which occur in the solid are $0 \leq x \leq 2$, and so its volume is

$$V = \int_0^2 \{3 - 4(1 - x)^2 + (1 - x)^4\} dx$$

$$\begin{aligned}
 &= \pi \left[3x + \frac{4}{3}(1-x)^3 - \frac{1}{5}(1-x)^5 \right]_0^2 \\
 &= \frac{55}{15}\pi
 \end{aligned}$$

4.8.4: Volumes by Cylindrical shells

Instead of slicing a solid with planes you can also try to decompose it into cylindrical shells. The volume of a cylinder of height h and radius r is $\pi r^2 h$ (height times area base). Therefore, the volume of a cylindrical shell of height h , (inner) radius r and thickness Δr is



$$\begin{aligned}
 \pi h(r + \Delta r)^2 - \pi h r^2 &= \pi h(2r + \Delta r)\Delta r \\
 &\approx 2\pi h r \Delta r
 \end{aligned}$$

Now consider the solid you get by revolving the region

$$R = \{(x, y) | a \leq x \leq b, 0 \leq y \leq f(x)\}$$

around the y -axis. By partitioning the interval $a \leq x \leq b$ into many small intervals we can decompose the solid into many thin shells. The volume of

each shell will approximately be given by $2\pi x f(x) \Delta x$. Adding the volumes of the shells, and taking the limit over finer and finer partitions we arrive at the following formula for the volume of the solid of revolutions:

$$V = 2\pi \int_a^b x f(x) dx \quad (41)$$

If the R is not the region under the graph, but rather the region between the graphs of two functions $f(x) \leq g(x)$, then we get

$$V = 2\pi \int_a^b x \{g(x) - f(x)\} dx$$

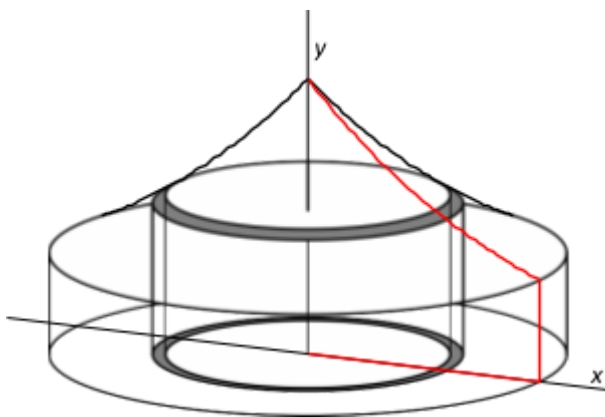


Figure 4.10: Computing the volume of a circus tent using cylindrical shells.

This particular tent is obtained by rotating the graph of $y = e^{-x}$, $0 \leq x \leq 1$, around the y - axis.

Example-The solid obtained by rotating R about the $y - axis$, again.

The R from (4.1) can also be described as

$$R = \{x, y | 0 \leq x \leq 2, f(x) \leq y \leq g(x)\}$$

where

$$f(x) = (x - 1)^2 \text{ and } g(x) = 1$$

The volume of the solid which we already computed in (4.1) is thus given by

$$\begin{aligned} V &= 2\pi \int_0^1 x \{1 - (x - 1)^2\} dx \\ &= 2\pi \int_0^2 -\{x^3 + 2x^2\} dx \\ &= 2\pi \left[-\frac{1}{4}x^4 + \frac{2}{3}x^3 \right]_0^2 \\ &= \frac{8\pi}{3} \end{aligned}$$

Which coincides with the answer we found in 4.1.

Exercises:

Group Problem

1. Find the volume enclosed by the paraboloid obtained by rotating the graph of $f(x) = R\sqrt{\frac{x}{H}}$ ($0 \leq x \leq H$) around the $x - axis$. Here R and H are positive constants.

Draw the solid whose volume you are asked to compute, and indicate what R and H are in your drawing. Find the volume of the solids you get by rotating each of the following graph around the $x - axis$

2. $f(x) = x, 0 \leq x \leq 2$
3. $f(x) = \sqrt{2 - x}, 0 \leq x \leq 2$
4. $f(x) = (1 + x^2)^{\frac{1}{2}}, |x| \leq 1$
5. $f(x) = \sin \sin x, 0 \leq x \leq \pi$
6. $f(x) = 1 - x^2, |x| \leq 1$
7. $f(x) = \cos x, 0 \leq x \leq \pi$
8. $f(x) = \frac{1}{\cos x}, 0 \leq x \leq \frac{\pi}{4}$
9. Find the volume that results by rotating the semicircle $y = \sqrt{R^2 - x^2}$ about the x - axis
10. Let T be triangle $1 \leq x \leq 2, 0 \leq y \leq 3x - 3$
 - a). find the volume of the solid obtained by rotating the triangle T around the x - axis
 - b). Find the volume that results by rotating the triangle T around the y - axis
 - c). Find the volume that results by rotating the triangle T around the line $x = -1$

Conclusion

In this session, we explored the methods of integration, particularly focusing on Riemann sums and the interpretation of integrals as the area under a curve. We discussed how to approximate the area using thin rectangles and defined

Riemann sums. The fundamental theorem of calculus was introduced, showing the relationship between antiderivatives and definite integrals. We also examined the concept of indefinite integrals and their notation.

Furthermore, we saw applications of integration, such as computing areas between graphs and volumes of solids of revolution. The method of slicing was used to find the volume of a pyramid and general solids of revolution, illustrating Cavalieri's principle. Finally, we applied integration to solve problems revolving around different axes, finding volumes of revolved solids. Integration emerges as a powerful tool for calculating areas, volumes, and solving various mathematical problems.

Summary

1. **Methods of Integration:** The integral of a function represents the area under the graph of the function on a specific interval. To approximate the area, the region is divided into thin rectangles using Riemann sums. The partition is made finer by increasing the number of division points, leading to a better approximation of the area. The integral of a positive function represents the area between the graph and the x-axis, while a negative function contributes negatively to the area.
2. **The Fundamental Theorem of Calculus:** An antiderivative of a function on an interval is called an antiderivative if its derivative matches the

original function. The Fundamental Theorem of Calculus states that if the integral of a function exists, and F is an antiderivative of that function on the interval, then the definite integral is equal to the difference of the antiderivative values at the endpoints. The bounds of the integral are referred to as " a " and " b ," the function being integrated is the "integrand," and " x " is the "integration variable." The integration variable is a dummy variable, and replacing it with another variable does not affect the result of the integral. Different antiderivatives of the same function differ by a constant.

3. The Indefinite Integral: The indefinite integral represents the antiderivative of a function and is denoted as $F(x)$. The integral of a function does not have specific integration bounds, unlike the definite integral. The indefinite integral is used when finding antiderivatives for complex functions. Some standard antiderivatives are used frequently to compute integrals of various functions.
4. Properties of the Integral: The integral exhibits similar properties to the limits and derivatives of sums and products of functions. For two functions $f(x)$ and $g(x)$, their antiderivatives are $F(x)$ and $G(x)$, respectively. The integral of the sum of two functions is the sum of their individual integrals. A constant can be factored out of the integral. The integral over an interval can be split into the sum of integrals over smaller

intervals. The definite integral can be used to calculate areas between graphs of two functions.

5. Method of Substitution: The chain rule helps compute integrals using substitution. The substitution method involves introducing new variables and finding their derivatives to simplify integrals. Substitution is useful when dealing with complex functions that are difficult to integrate directly. Substitution is also used for definite integrals, where the integration bounds are changed accordingly.
6. Applications of the Integral: The integral is used to calculate areas between graphs, especially when one function is always greater than the other. Cavalieri's principle states that if two solids have the same cross-sectional areas at any height, they have the same volume. The method of slicing is used to compute volumes of solids of revolution, where a function is rotated around an axis. Solids of revolution are created by rotating a positive function around an axis, and their volumes are computed using the integral.

References

1. Pankratov, L., Pettersson, I. and Podlipskaia (2021). Lecture Notes in Introductory Calculus. MOSCOW-MIPT.
2. Giusti, A. and Mainardi, F. (2020). Advanced Mathematical Methods: Theory and Applications. Multidisciplinary Digital Publishing Institute.
3. Strange, G. (2014). Calculus. Wellesley-Cambridge Press, Wellesley MA 02181
4. C. E. Chidume and C. O. Chidume (2012). Foundations of Mathematical Analysis; Ibadan University Press.

Glossary

Chain Rule	A rule used in calculus to find the derivative of a composite function.
Composition of a Continuous Function and a Limit	A theorem stating that if a function is continuous at a point and a limit exists at that point, then the composition of the two functions will also have a limit at that point.
Continuous Functions:	Functions that have no gaps, jumps, or holes in their graph.
Derivative	The rate at which a function's value changes concerning its independent variable.
Differentiability	The property of a function being differentiable, meaning its derivative exists at each point in its domain.
Differentiating	The process of finding the derivative of a function with respect to its independent variable.
Exponential Growth and Decay	A mathematical model used to describe the growth or decay of a quantity at a rate proportional to its current value.

Fundamental Theorem of Calculus	A theorem relating differentiation and integration, stating that the definite integral of a function can be evaluated using its antiderivative.
Indefinite Integral	An antiderivative of a function without specified integration bounds.
Limits	The value that a function approaches as its input approaches a particular value.
Polynomial Functions	Functions that consist of the sum of non-negative integer powers of a variable.
Product Rule	A rule used to find the derivative of the product of two functions.
Quotient Rule	A rule used to find the derivative of the quotient of two functions.
Rational Functions	Functions that are the ratio of two polynomials.
Sandwich Theorem	A theorem used to find the limit of a function between two other functions when they all approach the same value.

Substitution in Limits	A technique used to evaluate limits by substituting a variable with a specific value.
Sum, Product, and Quotient Rules:	Rules used to find the derivatives of functions involving sums, products, and quotients of other functions.
Trigonometric Functions	Functions based on trigonometric ratios such as sine, cosine, and tangent.